

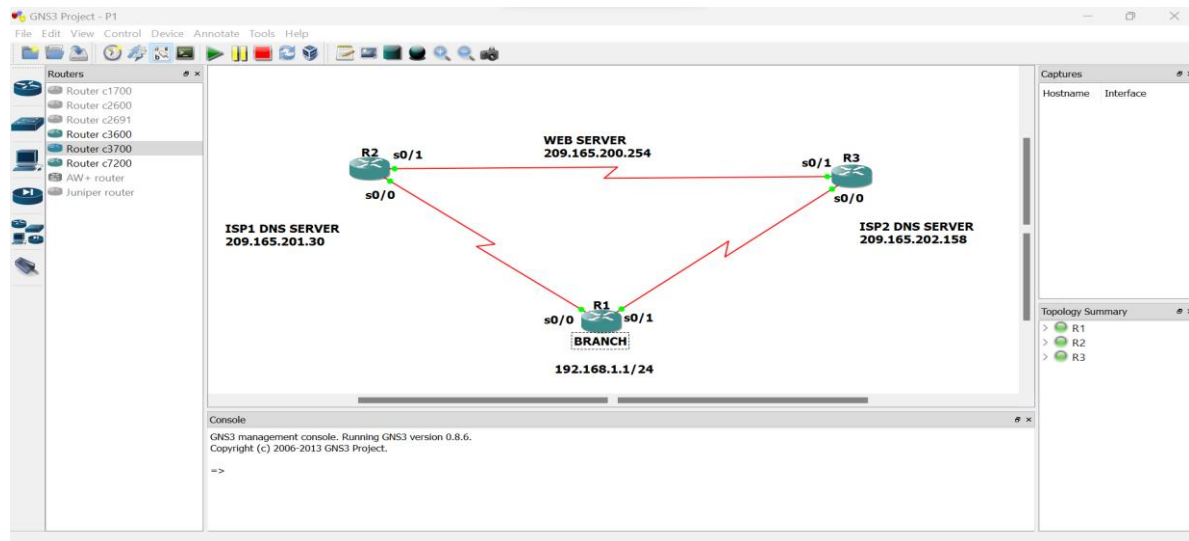
Modern Networking Practical's
Academic Year: 2022-2023

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PRACTICAL 1

Aim: Configure IP SLA Tracking and Path Control Topology

NETWORK TOPOLOGY



CONSOLE CODE

Step 1

R1 CONSOLE

```

Conf t
hostname R1
interface Loopback 0
description R1 LAN
ip address 192.168.1.1 255.255.255.0
interface Serial1/0
description R1 □ ISP1
ip address 209.165.201.2 255.255.255.252
clock rate 128000
bandwidth 128
no shutdown
interface Serial1/1
description R1 □ ISP2
ip address 209.165.202.130 255.255.255.252
bandwidth 128
no shutdown

```

Router ISP1 (R2) CONSOLE

```

Conf t
hostname ISP1
interface Loopback0
description Simulated Internet Web Server
ip address 209.165.200.254 255.255.255.255
interface Loopback1
description ISP1 DNS Server
ip address 209.165.201.30 255.255.255.255
interface Serial1/0
description ISP1 □ R1
ip address 209.165.201.1 255.255.255.252
bandwidth 128

```

```
no shutdown
interface Serial1/2
description ISP1 □ ISP2
ip address 209.165.200.225 255.255.255.252
clock rate 128000
bandwidth 128
no shutdown
```

Router ISP2 (R3) CONSOLE

```
conf t
hostname ISP2
interface Loopback0
description Simulated Internet Web Server
ip address 209.165.200.254 255.255.255.255
interface Loopback1
description ISP2 DNS Server
ip address 209.165.202.158 255.255.255.255
interface Serial1/1
description ISP2 □ R1
ip address 209.165.202.129 255.255.255.252
clock rate 128000
bandwidth 128
no shutdown
interface Serial1/2
description ISP2 □ ISP1
ip address 209.165.200.226 255.255.255.252
bandwidth 128
no shutdown
```

Step 2

R1 CONSOLE

```
end
show interfaces description | include up
```

```
R1#sh int description | include up
Se1/0                                up          up          R1 .. ISP1
Se1/1                                up          up          R1 .. ISP2
Lo0                                  up          up          R1 LAN
R1#
```

ISP1 CONSOLE

```
end
show interfaces description | include up
```

```
ISP1#show interfaces description | include up
Se1/0                                up          up          ISP1 .. R1
Se1/2                                up          up          ISP1 .. IS
P2
Lo0                                  up          up          Simulated
Internet Web Server
Lo1                                  up          up          ISP1 DNS S
erver
ISP1#
```

ISP2 CONSOLE

```
end
```

show interfaces description | include up

```
ISP2#show interfaces description | include up
Se1/1                up          up          ISP2 .. R1
Se1/2                up          up          ISP2 .. IS
P1
Lo0                  up          up          Simulated
Internet Web Server
Lo1                  up          up          ISP2 DNS S
erver
ISP2#
```

Step 3

R1 CONSOLE

```
conf t
ip route 0.0.0.0 0.0.0.0 209.165.201.1
```

ISP1 CONSOLE

```
Conf t
router eigrp 1
network 209.165.200.224 0.0.0.3
network 209.165.201.0 0.0.0.31
no auto-summary
ip route 192.168.1.0 255.255.255.0 209.165.201.2
```

ISP2 CONSOLE

```
conf t
router eigrp 1
network 209.165.200.224 0.0.0.3
network 209.165.202.128 0.0.0.31
no auto-summary
ip route 192.168.1.0 255.255.255.0 209.165.202.130
```

Step 4

ISP1 CONSOLE

```
exit
conf t
ip route 192.168.1.0 255.255.255.0 209.165.201.2
end
```

ISP2 CONSOLE

```
exit
conf t
ip route 192.168.1.0 255.255.255.0 209.165.201.2
end
```

Step 5

R1 CONSOLE

```
tclsh
foreach address {
209.165.200.254
209.165.201.30
209.165.202.158
} {
```

```
ping $address source 192.168.1.1
}
exit
```

```
R1(tcl)#tclsh

R1(tcl)#foreach address {
+>(tcl)#209.165.200.254
+>(tcl)#209.165.201.30
+>(tcl)#209.165.202.158
+>(tcl)#} {
+>(tcl)#ping $address source 192.168.1.1
+>(tcl)#}

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.254, timeout is 2 seconds:
s:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/44/60 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.201.30, timeout is 2 seconds:
s:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/36/56 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.202.158, timeout is 2 seconds:
s:
Packet sent with a source address of 192.168.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/72/104 ms
R1(tcl)#
```

```
Conf t
ip sla monitor 11
type echo protocol ipIcmpEcho 209.165.201.30
frequency 10
exit
ip sla monitor schedule 11 life forever start-time now
exit
show ip sla monitor configuration 11
```

```
R1(config)#exit

R1(tcl)#show ip sla monitor configuration 11
IP SLA Monitor, Infrastructure Engine-II.
Entry number: 11
Owner:
Tag:
Type of operation to perform: echo
Target address: 209.165.201.30
Request size (ARR data portion): 28
Operation timeout (milliseconds): 5000
Type Of Service parameters: 0x0
Verify data: No
Operation frequency (seconds): 10
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Number of history Lives kept: 0
Number of history Buckets kept: 15
--More--
*May 23 12:42:10.655: %SYS-5-CONFIG_I: Configured from console by console
History Filter Type: None
Enhanced History:

R1(tcl)#
```

```
show ip sla monitor statistics
```

```

R1(tcl)#
R1(tcl)#
R1(tcl)#sh ip sla monitor statistics
Round trip time (RTT)    Index 11
      Latest RTT: 40 ms
Latest operation start time: *12:45:40.643 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 22
Number of failures: 0
Operation time to live: Forever

```

```

Conf t
ip sla monitor 22
type echo protocol ipIcmpEcho 209.165.202.158
frequency 10
exit
ip sla monitor schedule 22 life forever start-time now
exit
show ip sla monitor configuration 22

```

```

R1(tcl)#show ip sla monitor configuration 22
IP SLA Monitor, Infrastructure Engine-II.
Entry number: 22
Owner:
Tag:
Type of operation to perform: echo
Target address: 209.165.202.158
Request size (ARR data portion): 28
Operation timeout (milliseconds): 5000
Type Of Service parameters: 0x0
Verify data: No
Operation frequency (seconds): 10
Next Scheduled Start Time: Start Time already passed
Group Scheduled : FALSE
Life (seconds): Forever
Entry Ageout (seconds): never
Recurring (Starting Everyday): FALSE
Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Number of statistic hours kept: 2
Number of statistic distribution buckets kept: 1
Statistic distribution interval (milliseconds): 20
Number of history Lives kept: 0
Number of history Buckets kept: 15
--More--
*May 23 12:48:06.459: %SYS-5-CONFIG_I: Configured from console by con
ole
History Filter Type: None
Enhanced History:
R1(tcl)#

```

```

show ip sla monitor statistics

```

```

R1(tcl)#show ip sla monitor statistics
Round trip time (RTT)    Index 11
      Latest RTT: 28 ms
Latest operation start time: *12:53:00.643 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 66
Number of failures: 0
Operation time to live: Forever

Round trip time (RTT)    Index 22
      Latest RTT: 76 ms
Latest operation start time: *12:52:56.443 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 30
Number of failures: 0
Operation time to live: Forever

```

Step 6

R1 CONSOLE

```

Conf t
no ip route 0.0.0.0 0.0.0.0 209.165.201.1
ip route 0.0.0.0 0.0.0.0 209.165.201.1 5
exit
show ip route

```

```

R1(tcl)#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

    209.165.201.0/30 is subnetted, 1 subnets
C       209.165.201.0 is directly connected, Serial1/0
    209.165.202.0/30 is subnetted, 1 subnets
C       209.165.202.128 is directly connected, Serial1/1
C       192.168.1.0/24 is directly connected, Loopback0
S*    0.0.0.0/0 [5/0] via 209.165.201.1
R1(tcl)#
*May 23 12:54:54.515: %SYS-5-CONFIG_I: Configured from console by console
R1(tcl)#

```

Step 7

R1 CONSOLE

```

conf t
track 1 rtr 11 reachability
delay down 10 up 1
exit
exit
debug ip routing
conf t
ip route 0.0.0.0 0.0.0.0 209.165.201.1 2 track 1

```

```

conf t
track 2 rtr 22 reachability
delay down 10 up 1
exit
exit
debug ip routing
conf t
ip route 0.0.0.0 0.0.0.0 209.165.202.129 3 track 2
exit
show ip route

```

```

R1(tcl)#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

    209.165.201.0/30 is subnetted, 1 subnets
C       209.165.201.0 is directly connected, Serial1/0
    209.165.202.0/30 is subnetted, 1 subnets
C       209.165.202.128 is directly connected, Serial1/1
C       192.168.1.0/24 is directly connected, Loopback0
S*    0.0.0.0/0 [2/0] via 209.165.201.1
*May 23 12:56:52.475: %SYS-5-CONFIG_I: Configured from console by console
*May 23 12:56:52.591: RT: closer admin distance for 0.0.0.0, flushing 1 routes
*May 23 12:56:52.591: RT: NET-RED 0.0.0.0/0
*May 23 12:56:52.591: RT: SET_LAST_RDB for 0.0.0.0/0
    NEW rdb: via 209.165.201.1
*May 23 12:56:52.595: RT: add 0.0.0.0/0 via 209.165.201.1, static metric 5

```

Step 8

ISP1 CONSOLE

```

conf t
interface loopback 1
shutdown

```

R1 CONSOLE

show ip route

```

R1(tcl)#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 209.165.202.129 to network 0.0.0.0

    209.165.201.0/30 is subnetted, 1 subnets
C       209.165.201.0 is directly connected, Serial1/0
    209.165.202.0/30 is subnetted, 1 subnets
C       209.165.202.128 is directly connected, Serial1/1
C       192.168.1.0/24 is directly connected, Loopback0
S*     0.0.0.0/0 [3/0] via 209.165.202.129
R1(tcl)#

```

show ip sla monitor statistics

```

R1(tcl)#show ip sla monitor statistics
Round trip time (RTT)   Index 11
    Latest RTT: NoConnection/Busy/Timeout
Latest operation start time: *13:07:20.643 UTC Tue May 23 2023
Latest operation return code: Timeout
Number of successes: 142
Number of failures: 10
Operation time to live: Forever

Round trip time (RTT)   Index 22
    Latest RTT: 44 ms
Latest operation start time: *13:07:16.443 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 116
Number of failures: 0
Operation time to live: Forever

```

trace 209.165.200.254 source 192.168.1.1

```

R1(tcl)#trace 209.165.200.254 source 192.168.1.1

Type escape sequence to abort.
Tracing the route to 209.165.200.254

  0  1 209.165.202.129 48 msec 68 msec 52 msec
R1(tcl)#
*May 23 13:09:25.923: RT: NET-RED 0.0.0.0/0
R1(tcl)#

```

ISP1 CONSOLE

no shutdown

R1 CONSOLE

show ip sla monitor statistics

```

R1(tcl)#sh ip sla monitor statistics
Round trip time (RTT)   Index 11
    Latest RTT: 60 ms
Latest operation start time: *13:13:00.643 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 146
Number of failures: 40
Operation time to live: Forever

Round trip time (RTT)   Index 22
    Latest RTT: 84 ms
Latest operation start time: *13:12:56.443 UTC Tue May 23 2023
Latest operation return code: OK
Number of successes: 150
Number of failures: 0
Operation time to live: Forever

```

Show ip route

```

R1(tcl)#Show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 209.165.201.1 to network 0.0.0.0

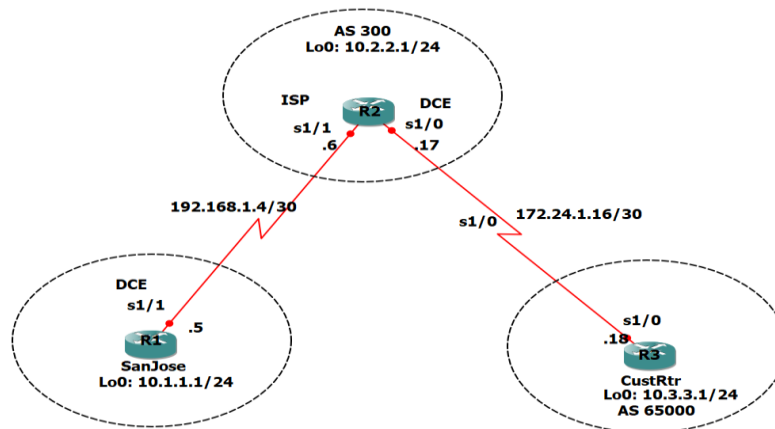
    209.165.201.0/30 is subnetted, 1 subnets
C       209.165.201.0 is directly connected, Serial1/0
    209.165.202.0/30 is subnetted, 1 subnets
C       209.165.202.128 is directly connected, Serial1/1
C       192.168.1.0/24 is directly connected, Loopback0
S*     0.0.0.0/0 [2/0] via 209.165.201.1
R1(tcl)#

```


PRACTICAL 2

Aim: Using the AS_PATH Attribute.

Topology:



Step 1: Prepare the routers for the lab.

Cable the network as shown in the topology diagram. Erase the startup configuration and reload each router to clear previous configurations.

Step 2: Configure the hostname and interface addresses.

a. You can copy and paste the following configurations into your routers to begin.

Router R1 (hostname SanJose)

```

hostname SanJose
interface Loopback0
ip address 10.1.1.1 255.255.255.0
interface Serial1/1
ip address 192.168.1.5 255.255.255.252
clock rate 128000
no shutdown
  
```

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname SanJose
SanJose(config)#interface Loopback0
SanJose(config-if)#ip address 10.1.1.1 255.255.255.0
SanJose(config-if)#interface Serial1/1
SanJose(config-if)#ip address 192.168.1.5 255.255.255.252
SanJose(config-if)#clock rate 128000
SanJose(config-if)#no shutdown
  
```

Router R2 (hostname ISP)

```

hostname ISP
interface Loopback0
ip address 10.2.2.1 255.255.255.0
interface Serial1/1
ip address 192.168.1.6 255.255.255.252
no shutdown
interface Serial1/0
ip address 172.24.1.17 255.255.255.252
clock rate 128000
no shutdown
  
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname ISP
ISP(config)#interface Loopback0
ISP(config-if)#ip address 10.2.2.1 255.255.255.0
ISP(config-if)#interface Serial1/1
ISP(config-if)#ip address 192.168.1.6 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#interface Serial1/0
ISP(config-if)#ip address 172.24.1.17 255.255.255.252
ISP(config-if)#clock rate 128000
```

Router R3 (hostname CustRtr)

```
hostname CustRtr
interface Loopback0
ip address 10.3.3.1 255.255.255.0
interface Serial1/0
ip address 172.24.1.18 255.255.255.252
no shutdown
```

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname CustRtr
CustRtr(config)#interface Loopback0
CustRtr(config-if)#ip address 10.3.3.1 255.255.255.0
CustRtr(config-if)#interface Serial1/0
CustRtr(config-if)#ip address 172.24.1.18 255.255.255.252
CustRtr(config-if)#no shutdown
```

b. Use **ping** to test the connectivity between the directly connected routers.

Note: SanJose will not be able to reach either ISP's loopback (10.2.2.1) or CustRtr's loopback (10.3.3.1), nor will it be able to reach either end of the link joining ISP to CustRtr (172.24.1.17 and 172.24.1.18).

Step 3: Configure BGP.

a. Configure BGP for normal operation. Enter the appropriate BGP commands on each router so that they identify their BGP neighbors and advertise their loopback networks.

Router R1 (hostname SanJose)

```
router bgp 100
neighbor 192.168.1.6 remote-as 300
network 10.1.1.0 mask 255.255.255.0
```

```
SanJose(config)#router bgp 100
SanJose(config-router)#neighbor 192.168.1.6 remote-as 300
SanJose(config-router)#network 10.1.1.0 mask 255.255.255.0
SanJose(config-router)#
```

Router R2 (hostname ISP)

```
router bgp 300
neighbor 192.168.1.5 remote-as 100
neighbor 172.24.1.18 remote-as 65000
network 10.2.2.0 mask 255.255.255.0
```

```
ISP(config-if)#router bgp 300
ISP(config-router)#neighbor 192.168.1.5 remote-as 100
ISP(config-router)#neighbor 172.24.1.18 remote-as 65000
ISP(config-router)#
*Jun 21 13:21:43.239: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
ISP(config-router)#network 10.2.2.0 mask 255.255.255.0
ISP(config-router)#
```

Router R3 (hostname CustRtr)

```
router bgp 65000
neighbor 172.24.1.17 remote-as 300
network 10.3.3.0 mask 255.255.255.0
```

```
CustRtr(config-if)#router bgp 65000
CustRtr(config-router)#neighbor 172.24.1.17 remote-as 300
CustRtr(config-router)#network 10.3.3.0 mask 255.255.255.0
CustRtr(config-router)#
```

b. Verify that these routers have established the appropriate neighbor relationships by issuing the **show ip bgp neighbors** command on each router.

Router R2 (hostname ISP)

show ip bgp neighbors

```

R2
Connection is ECN Disabled, Minimum incoming TTL 0, Outgoing TTL 1
Local host: 192.168.1.6, Local port: 179
Foreign host: 192.168.1.5, Foreign port: 35468

Enqueued packets for retransmit: 0, input: 0  mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x1F4124):
Timer           Starts      Wakeups      Next
Retrans         11           0            0x0
TimeWait        0            0            0x0
AckHold         12           3            0x0
SendWnd         0            0            0x0
KeepAlive       0            0            0x0
GiveUp          0            0            0x0
PmtuAger        0            0            0x0
DeadWait        0            0            0x0

iss: 1283501914  snduna: 1283502249  sndnxt: 1283502249      sndwnd: 16050
rs: 3703581546  rcvnxt: 3703581834  rcvwnd: 16097  delrcvwnd: 287

BRTT: 231 ms, RTTO: 769 ms, RTV: 538 ms, KRTT: 0 ms
minRTT: 32 ms, maxRTT: 300 ms, ACK hold: 200 ms
Flags: passive open, nagle, gen tcbs
IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):
Rcvd: 21 (out of order: 0), with data: 12, total data bytes: 287
Sent: 15 (retransmit: 0, fastretransmit: 0, partialack: 0, Second Congestion: 0)
Packets: 334
ISP#

```

Step 4: Remove the private AS.

a. Display the SanJose routing table using the **show ip route** command. SanJose should have a route to both 10.2.2.0 and 10.3.3.0. Troubleshoot if necessary.

Router R1 (hostname SanJose)

show ip route

```

SanJose#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 2 subnets
B       10.2.2.0 [20/0] via 192.168.1.6, 00:12:00
C       10.1.1.0 is directly connected, Loopback0
       192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.4 is directly connected, Serial1/1
SanJose#

```

c. Ping again, this time as an extended ping, sourcing from the Loopback0 interface address.

Router R1 (hostname SanJose)

SanJose#ping

Protocol [ip]:

Target IP address: **10.3.3.1**

Repeat count [5]:

Datagram size [100]:

Timeout in seconds [2]:

Extended commands [n]: y

Source address or interface: **10.1.1.1**

Type of service [0]:

Set DF bit in IP header? [no]:
 Validate reply data? [no]:
 Data pattern [0xABCD]:
 Loose, Strict, Record, Timestamp, Verbose[none]:
 Sweep range of sizes [n]:
 Type escape sequence to abort.
 Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
 !!!!
 Success rate is 100 percent (5/5), round-trip min/avg/max = 64/64/68 ms
Note: You can bypass extended ping mode and specify a source address using one of these commands:
SanJose# ping 10.3.3.1 source 10.1.1.1

d. Check the BGP table from SanJose by using the **show ip bgp** command. Note the AS path for the 10.3.3.0 network. The AS 65000 should be listed in the path to 10.3.3.0.

Router R1 (hostname SanJose)

show ip bgp

```
SanJose#sh ip bgp
BGP table version is 3, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 10.1.1.0/24      0.0.0.0              0         32768 i
*> 10.2.2.0/24      192.168.1.6          0           0 300 i
SanJose#
```

e. Configure ISP to strip the private AS numbers from BGP routes exchanged with SanJose using the following commands.

Router R2 (hostname ISP)

router bgp 300

neighbor 192.168.1.5 remove-private-as

```
ISP#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP(config)#router bgp 300
ISP(config-router)#neighbor 192.168.1.5 remove-private-as
ISP(config-router)#
```

F. After issuing these commands, use the **clear ip bgp *** command on **ISP** to reestablish the BGP relationship between the three routers. Wait several seconds and then return to SanJose to check its routing table.

Note: The **clear ip bgp * soft** command can also be used to force each router to resend its BGP table.

Does SanJose still have a route to 10.3.3.0?

```
ISP#clear ip bgp *
ISP#
*Jun 21 14:05:31.947: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
```

SanJose should be able to ping 10.3.3.1 using its loopback 0 interface as the source of the ping.

Router R1 (hostname SanJose)

ping 10.3.3.1 source lo0

```
SanJose#ping 10.3.3.1 source lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
.....
Success rate is 0 percent (0/5)
```

G. Now check the BGP table on SanJose. The AS_PATH to the 10.3.3.0 network should be AS 300. It no longer has the private AS in the path.

Router R1 (hostname SanJose)

show ip bgp

```
SanJose#sh ip bgp
BGP table version is 5, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 10.1.1.0/24      0.0.0.0              0         32768 i
*> 10.2.2.0/24      192.168.1.6          0           0 300 i
SanJose#
```

Step 5: Use the AS_PATH attribute to filter routes.

As a final configuration, use the AS_PATH attribute to filter routes based on their origin. In a complex environment, you can use this attribute to enforce routing policy. In this case, the provider router, ISP, must be configured so that it does not propagate routes that originate from AS 100 to the customer router CustRtr. \

a. Configure a special kind of access list to match BGP routes with an AS_PATH attribute that both begins and ends with the number 100. Enter the following commands on ISP.

Router R2 (hostname ISP)

```
ip as-path access-list 1 deny ^100$
```

```
ip as-path access-list 1 permit .*
```

```
ISP(config)#ip as-path access-list 1 deny ^100$
ISP(config)#ip as-path access-list 1 permit .*
ISP(config)#
```

The first command uses the ^ character to indicate that the AS path must begin with the given number 100. The \$ character indicates that the AS_PATH attribute must also end with 100. Essentially, this statement matches only paths that are sourced from AS 100. Other paths, which might include AS 100 along the way, will not match this list.

In the second statement, the . (period) is a wildcard, and the * (asterisk) stands for a repetition of the wildcard. Together, .* matches any value of the AS_PATH attribute, which in effect permits any update that has not been denied by the previous **access-list** statement.

For more details on configuring regular expressions on Cisco routers, see:

http://www.cisco.com/en/US/docs/ios/12_2/termserv/configuration/guide/tcfaapre_ps1835_TSD_Products_Configuration_Guide_Chapter.html

B. Apply the configured access list using the **neighbor** command with the **filter-list** option.

Router R2 (hostname ISP)

```
router bgp 300
```

```
neighbor 172.24.1.18 filter-list 1 out
```

```
ISP(config)#router bgp 300
ISP(config-router)#neighbor 172.24.1.18 filter-list 1 out
ISP(config-router)#
```

The **out** keyword specifies that the list is applied to routing information sent to this neighbor.

C. Use the **clear ip bgp *** command to reset the routing information. Wait several seconds and then check the routing table for **ISP**. The route to 10.1.1.0 should be in the routing table.

```
ISP#clear ip bgp *
ISP#
*Jun 21 14:25:40.879: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
ISP#
*Jun 21 14:25:43.247: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
```

Note: To force the local router to resend its BGP table, a less disruptive option is to use the **clear ip bgp * out** or **clear ip bgp * soft** command (the second command performs both outgoing and incoming route resync).

Router R2 (hostname ISP)

```
show ip route
```

```
ISP#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 2 subnets
C    10.2.2.0 is directly connected, Loopback0
B    10.1.1.0 [20/0] via 192.168.1.5, 00:01:02
     192.168.1.0/30 is subnetted, 1 subnets
C    192.168.1.4 is directly connected, Serial1/1
ISP#
```

D. Check the routing table for CustRtr. It should not have a route to 10.1.1.0 in its routing table.

Router R3 (hostname CustRtr)

show ip route

```
CustRtr#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets
C    10.3.3.0 is directly connected, Loopback0
CustRtr#
```

E. Return to ISP and verify that the filter is working as intended. Issue the **show ip bgp regexp ^100\$** command.

Router R2 (hostname ISP)

show ip bgp regexp ^100\$

```
ISP#sh ip bgp regexp ^100$
BGP table version is 3, local router ID is 10.2.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
*> 10.1.1.0/24      192.168.1.5          0           0 100 i
ISP#
```

The output of this command shows all matches for the regular expressions that were used in the access list. The path to 10.1.1.0 matches the access list and is filtered from updates to CustRtr.

F. Run the following Tcl script on all routers to verify whether there is connectivity. All pings from ISP should be successful. SanJose should not be able to ping the CustRtr loopback 10.3.3.1 or the WAN link 172.24.1.16/30. CustRtr should not be able to ping the SanJose loopback 10.1.1.1 or the WAN link 192.168.1.4/30.

Router R2 (hostname ISP)

ISP#tclsh

ISP(tcl)#foreach address {

+>10.1.1.1

+>10.2.2.1

+>10.3.3.1

+>192.168.1.5

+>192.168.1.6

+>172.24.1.17

+>172.24.1.18 }

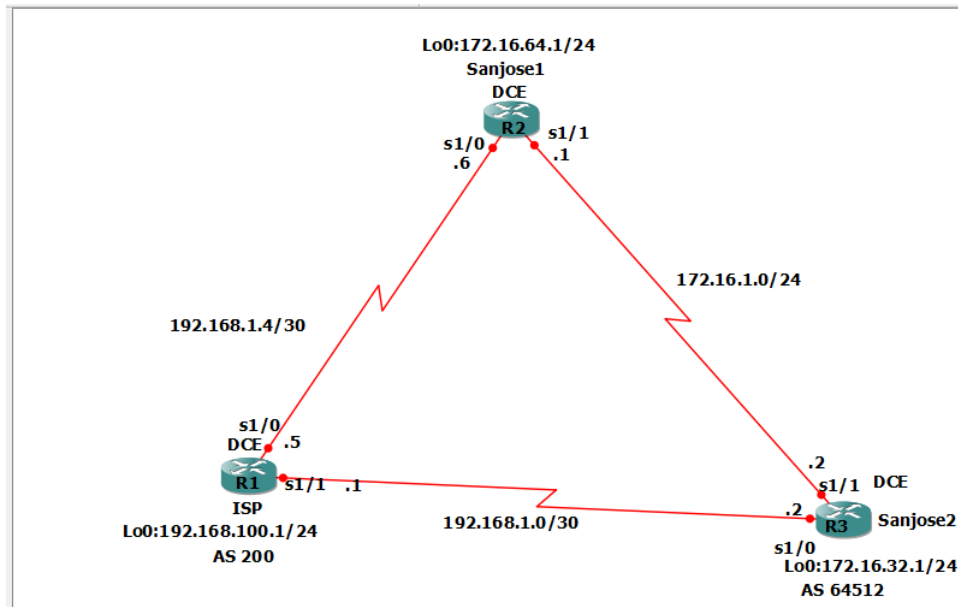
Ping \$address

Router Interface Summary				
Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1700	Fast Ethernet 0 (FA0)	Fast Ethernet 1 (FA1)	Serial 0 (S0)	Serial 1 (S1)
1800	Fast Ethernet 0/0 (FA0/0)	Fast Ethernet 0/1 (FA0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2600	Fast Ethernet 0/0 (FA0/0)	Fast Ethernet 0/1 (FA0/1)	Serial 0/0 (S0/0)	Serial 0/1 (S0/1)
2800	Fast Ethernet 0/0 (FA0/0)	Fast Ethernet 0/1 (FA0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. Rather than list all combinations of configurations for each router class, this table includes identifiers for the possible combinations of Ethernet and serial interfaces in the device. The table does not include any other type of interface, even though a specific router might contain one. For example, for an ISDN BRI interface, the string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.				

PRACTICAL 3

AIM: Configuring IBGP and EBGP Sessions, Local Preference, and MED

Topology:



CODE:

Step 1: Configure interface addresses.

- Using the addressing scheme in the diagram, create the loopback interfaces and apply IPv4 addresses to these and the serial interfaces on ISP (R1), SanJose1 (R2), and SanJose2 (R3). Apply the following configuration to each router along with the appropriate hostname.

R1 (hostname ISP)

```
conf t
hostname ISP
```

R2 (hostname SanJose1)

```
conf t
hostname SanJose1
```

R3(hostname SanJose2)

```
conf t
hostname SanJose2
```

Run the following in all the router consoles:

```
no ip domain-lookup
line con 0
logging synchronous
exec-timeout 0 0
```

R1(hostname ISP)

```
conf t
```



```
hostname ISP
interface Loopback0
ip address 192.168.100.1 255.255.255.0
exit
interface S1/0
ip address 192.168.1.5 255.255.255.252
clock rate 128000
no shutdown
exit
interface S1/1
ip address 192.168.1.1 255.255.255.252
no shutdown
end
```

R2 (hostname SanJose1)

```
conf t
hostname SanJose1
interface Loopback0
ip address 192.168.100.1 255.255.255.0
exit
interface S1/0
ip address 192.168.1.5 255.255.255.252
clock rate 128000
no shutdown
exit
interface S1/1
ip address 192.168.1.1 255.255.255.252
no shutdown
End
```

Step 2: Configure EIGRP.

Configure EIGRP between the SanJose1 and SanJose2 routers. (Note: If using an IOS prior to 15.0, use the no auto-summary router configuration command to disable automatic summarization. This command is the default **beginning with IOS 15.**)

Step 3: Configure IBGP and verify BGP neighbors.

- a. Configure IBGP between the SanJose1 and SanJose2 routers.

R2(hostname SanJose1)

```
router bgp 64512
neighbor 172.16.32.1 remote-as 64512
neighbor 172.16.32.1 update-source lo0
```

- b. Complete the IBGP configuration on SanJose2 using the following commands.

R3(hostname SanJose2)

```
router bgp 64512
neighbor 172.16.64.1 remote-as 64512
neighbor 172.16.64.1 update-source lo0
```

c. Verify that SanJose1 and SanJose2 become BGP neighbors by issuing the show ip bgp neighbors command on SanJose1.

View the following partial output. If the BGP state is not established, troubleshoot the connection.

exit

exit

SanJose2# show ip bgp neighbors

```
SanJose1#sh ip bgp neighbors
BGP neighbor is 172.16.32.1, remote AS 64512, internal link
BGP version 4, remote router ID 0.0.0.0
BGP state = Active
Last read 00:15:17, last write 00:15:17, hold time is 180,
keepalive interval is 60 seconds
Message statistics:
  InQ depth is 0
  OutQ depth is 0

      Sent      Rcvd
Opens:          0          0
Notifications:  0          0
Updates:        0          0
Keepalives:     0          0
Route Refresh:  0          0
Total:          0          0
Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
BGP table version 1, neighbor version 0/0
Output queue size : 0
Index 1, Offset 0, Mask 0x2
1 update-group member

      Sent      Rcvd
Prefix activity:
  Prefixes Current:    0          0
  Prefixes Total:      0          0
  Implicit Withdraw:   0          0
  Explicit Withdraw:   0          0
  Used as bestpath:    n/a          0
  Used as multipath:    n/a          0

      Outbound    Inbound
Local Policy Denied Prefixes:
  Total:          0          0
Number of NLRI in the update sent: max 0, min 0

Connections established 0; dropped 0
Last reset never
No active TCP connection
```

Step 4: Configure EBGW and verify BGP neighbours.

a. Configure ISP to run EBGW with SanJose1 and SanJose2. Enter the following commands on ISP.

R1(hostname ISP)

ISP(config)# router bgp 200

ISP(config-router)# neighbor 192.168.1.6 remote-as 64512

ISP(config-router)# neighbor 192.168.1.2 remote-as 64512

ISP(config-router)# network 192.168.100.0

b. Configure a discard static route for the 172.16.0.0/16 network. Any packets that do not have a more specific match (longer match) for a 172.16.0.0 subnet will be dropped instead of sent to the ISP. Later in this lab we will configure a default route to the ISP.

R2(hostname SanJose1)

exit

SanJose1(config)# ip route 172.16.0.0 255.255.0.0 null0

c. Configure SanJose1 as an EBGW peer to ISP.

R2(hostname SanJose1)

```
SanJose1(config)# router bgp 64512
SanJose1(config-router)# neighbor 192.168.1.5 remote-as 200
SanJose1(config-router)# network 172.16.0.0
```

d. Use the show ip bgp neighbors command to verify that SanJose1 and ISP have reached the established state. Troubleshoot if necessary.

R2(hostname SanJose1)

```
SanJose1# show ip bgp neighbors
```

```
SanJose1#show ip bgp neighbors
BGP neighbor is 172.16.32.1, remote AS 64512, internal link
  BGP version 4, remote router ID 0.0.0.0
  BGP state = Active
  Last read 00:45:07, last write 00:45:07, hold time is 180, keepalive interval is 60 seconds
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
Opens:          0        0
Notifications:  0        0
Updates:        0        0
Keepalives:     0        0
Route Refresh:  0        0
Total:          0        0
  Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
  BGP table version 2, neighbor version 0/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member

      Sent      Rcvd
Prefix activity:  ----  ----
  Prefixes Current:    0        0
  Prefixes Total:      0        0
  Implicit Withdraw:   0        0
  Explicit Withdraw:   0        0
  Used as bestpath:    n/a        0
  Used as multipath:    n/a        0

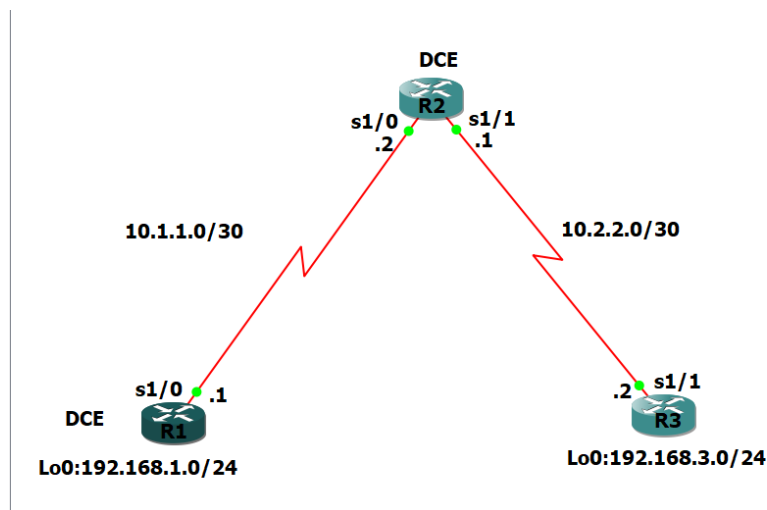
      Outbound   Inbound
Local Policy Denied Prefixes:  -----  -----
  Total:                      0        0
  Number of NLRI in the update sent: max 0, min 0

Connections established 0; dropped 0
Last reset never
No active TCP connection
```

PRACTICAL 4

AIM: Secure the Management Plane

Topology:



Step 1: Configure loopbacks and assign addresses.

R1 Console

```
interface Loopback 0
ip address 192.168.1.1 255.255.255.0
exit
```

R1 Console

```
conf t
interface Serial1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

R2 Console

```
interface Serial1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
```

R2 Console

```
conf t
interface Serial1/1
ip address 10.2.2.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

R3 Console

```
interface Loopback0
ip address 192.168.3.1 255.255.255.0
exit
```

R3 Console

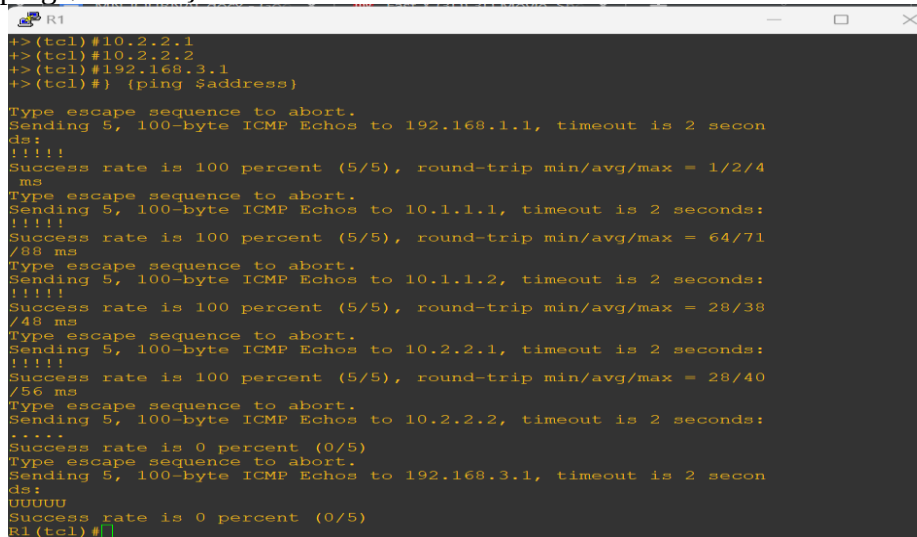
```
interface Serial1/1
ip address 10.2.2.2 255.255.255.252
no shutdown
exit
```

Step 2: Configure static routes.**R1 Console**

```
ip route 0.0.0.0 0.0.0.0 10.1.1.2
```

R1 Console

```
R1#tclsh
foreach address {
+>(tcl)#192.168.1.1
+>(tcl)#10.1.1.1
+>(tcl)#10.1.1.2
+>(tcl)#10.2.2.1
+>(tcl)#10.2.2.2
+>(tcl)#192.168.3.1
+>(tcl)#} { ping $address }
```



```
R1
+>(tcl)#10.2.2.1
+>(tcl)#10.2.2.2
+>(tcl)#192.168.3.1
+>(tcl)#} {ping $address}

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/71/88 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/38/48 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/40/56 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
R1(tcl)#
```

R2 Console

```
ip route 192.168.1.0 255.255.255.0 10.1.1.1
```

```
ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

```
R2(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)#ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

R3 Console

```
ip route 0.0.0.0 0.0.0.0 10.2.2.1
```

Step 3: Secure management access.**R1 Console**

```
security passwords min-length 10
```

```
enable secret class12345
```

```
line console 0
```

```
password ciscoconpass
```

```
exec-timeout 5 0
```

```
login
```

```
logging synchronous
```

```
exit
```

```
line vty 0 4
```

```
password ciscovtypass
```

```
exec-timeout 5 0
```

```
login
```

```
exit
```

```
line aux 0
no exec
end
```

```
service password-encryption
banner motd $Unauthorized access strictly prohibited!$
exit
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#service password-encryption
R1(config)#banner motd $Unauthorized access strickly prohibited!$
R1(config)#exit
R1#
*May 14 12:47:10.547: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3 Console

```
security passwords min-length 10
enable secret class12345
line console 0
password ciscoconpass
exec-timeout 5 0
login
logging synchronous
exit
```

```
line vty 0 4
password ciscovtypass
exec-timeout 5 0
login
exit
```

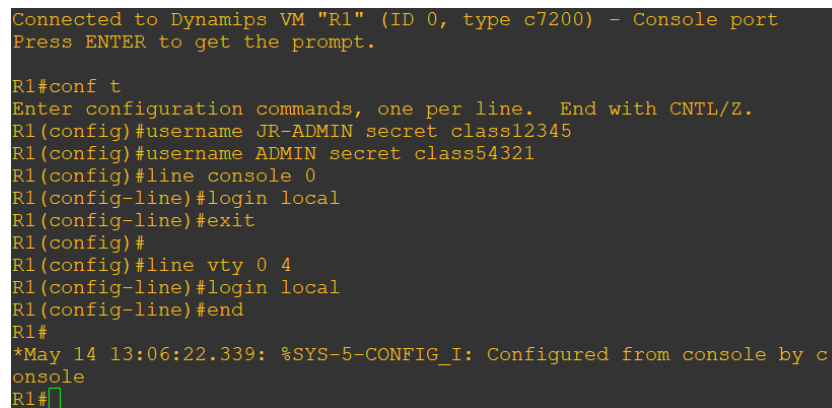
```
line aux 0
no exec
end
```

```
service password-encryption
banner motd $Unauthorized access strictly prohibited!$
exit
```

```
R3(config-line)#logging synchronous
R3(config-line)#exit
R3(config)#
R3(config)#line vty 0 4
R3(config-line)#password ciscovtypass
R3(config-line)#exec-timeout 5 0
R3(config-line)#login
R3(config-line)#exit
R3(config)#
R3(config)#line aux 0
R3(config-line)#no exec
R3(config-line)#end
R3#
*May 14 13:00:35.807: %SYS-5-CONFIG_I: Configured from console by console
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#service password-encryption
R3(config)#banner motd $Unauthorized-access strictly prohibited!$
R3(config)#exit
R3#
*May 14 13:02:55.295: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Step 4: Configure enhanced username password security.**R1 Console**

```
username JR-ADMIN secret class12345
username ADMIN secret class54321
line console 0
login local
exit
line vty 0 4
login local
end
```

A screenshot of a terminal window titled "Connected to Dynamips VM 'R1' (ID 0, type c7200) - Console port". The prompt is "Press ENTER to get the prompt." The user enters "R1#conf t", followed by "Enter configuration commands, one per line. End with CNTL/Z." The configuration commands entered are: "R1(config)#username JR-ADMIN secret class12345", "R1(config)#username ADMIN secret class54321", "R1(config)#line console 0", "R1(config-line)#login local", "R1(config-line)#exit", "R1(config)#", "R1(config)#line vty 0 4", "R1(config-line)#login local", "R1(config-line)#end", and "R1#". A system message follows: "*May 14 13:06:22.339: %SYS-5-CONFIG_I: Configured from console by console". The prompt "R1#" is visible at the bottom.

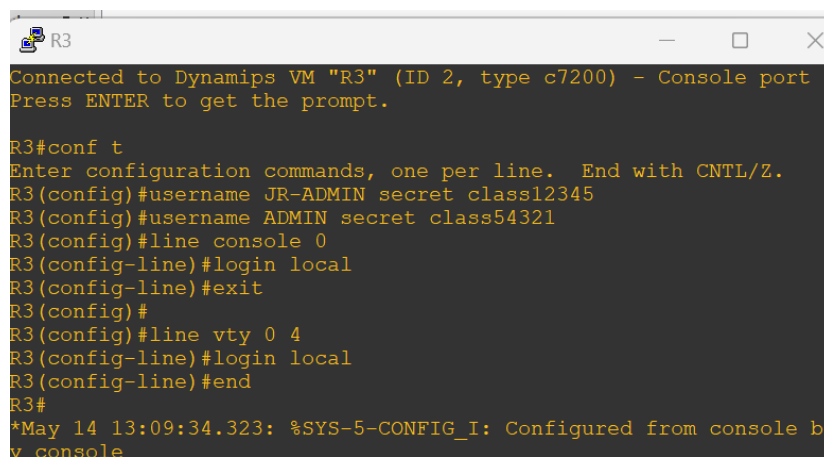
```
Connected to Dynamips VM "R1" (ID 0, type c7200) - Console port
Press ENTER to get the prompt.

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#username JR-ADMIN secret class12345
R1(config)#username ADMIN secret class54321
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#exit
R1(config)#
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#end
R1#
*May 14 13:06:22.339: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3 Console

```
username JR-ADMIN secret class12345
username ADMIN secret class54321
line console 0
login local
exit

line vty 0 4
login local
end
```

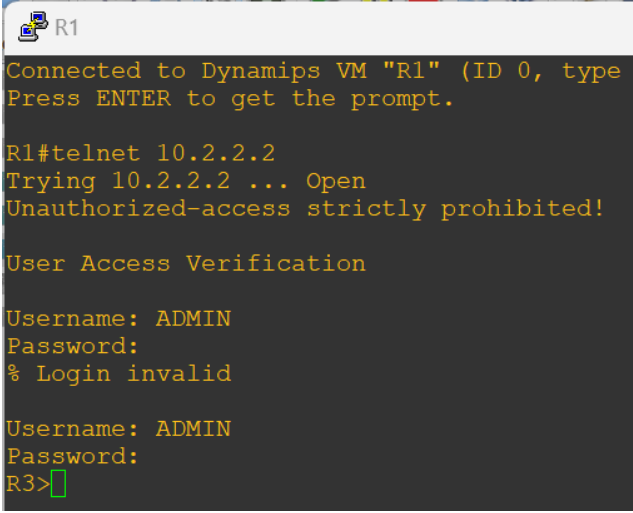
A screenshot of a terminal window titled "Connected to Dynamips VM 'R3' (ID 2, type c7200) - Console port". The prompt is "Press ENTER to get the prompt." The user enters "R3#conf t", followed by "Enter configuration commands, one per line. End with CNTL/Z." The configuration commands entered are: "R3(config)#username JR-ADMIN secret class12345", "R3(config)#username ADMIN secret class54321", "R3(config)#line console 0", "R3(config-line)#login local", "R3(config-line)#exit", "R3(config)#", "R3(config)#line vty 0 4", "R3(config-line)#login local", "R3(config-line)#end", and "R3#". A system message follows: "*May 14 13:09:34.323: %SYS-5-CONFIG_I: Configured from console by console". The prompt "R3#" is visible at the bottom.

```
Connected to Dynamips VM "R3" (ID 2, type c7200) - Console port
Press ENTER to get the prompt.

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#username JR-ADMIN secret class12345
R3(config)#username ADMIN secret class54321
R3(config)#line console 0
R3(config-line)#login local
R3(config-line)#exit
R3(config)#
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#end
R3#
*May 14 13:09:34.323: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

R1 Console

```
telnet 10.2.2.2
```



```
R1
Connected to Dynamips VM "R1" (ID 0, type
Press ENTER to get the prompt.

R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
Unauthorized-access strictly prohibited!

User Access Verification

Username: ADMIN
Password:
% Login invalid

Username: ADMIN
Password:
R3>
```

Step 5: Enabling AAA RADIUS Authentication with Local User for Backup.

R1 Console

```
aaa new-model
radius server RADIUS-1
address ipv4 192.168.1.101
key RADIUS-1-pa55w0rd
exit
```

```
radius server RADIUS-2
address ipv4 192.168.1.102
key RADIUS-2-pa55w0rd
exit
```

```
aaa group server radius RADIUS-GROUP
server name RADIUS-1
server name RADIUS-2
exit
```

```
aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-case
line vty 0 4
login authentication TELNET-LOGIN
exit
```

R3 Console

```
aaa new-model
radius server RADIUS-1
address ipv4 192.168.1.101
key RADIUS-1-pa55w0rd
exit
```

```
radius server RADIUS-2
address ipv4 192.168.1.102
key RADIUS-2-pa55w0rd
exit
```

```
aaa group server radius RADIUS-GROUP
server name RADIUS-1
```

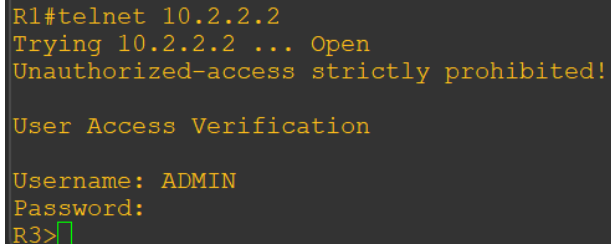


```
server name RADIUS-2
exit
```

```
aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-case
line vty 0 4
login authentication TELNET-LOGIN
exit
```

R1 Console

```
telnet 10.2.2.2
```



```
R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
Unauthorized-access strictly prohibited!

User Access Verification

Username: ADMIN
Password:
R3>
```

Step 6: Enabling secure remote management using SSH.

R1 Console

```
ip domain-name ccnasecurity.com
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line vty 0 4
transport input ssh
end
```

R3 Console

```
ip domain-name ccnasecurity.com
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
line vty 0 4
transport input ssh
end
```

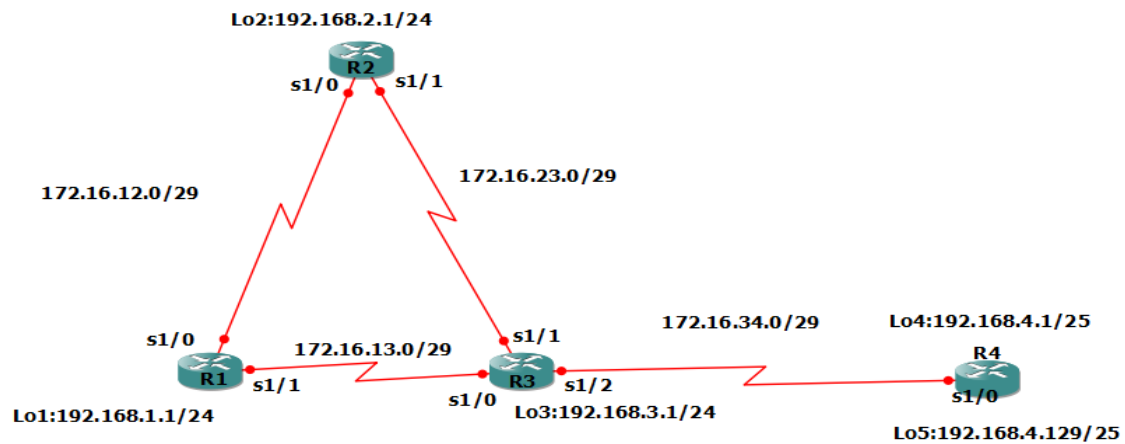
R1 Console

```
ssh -l ADMIN 10.2.2.2
```

PRACTICAL 5

AIM: Configure and Verify Path Control Using PBR.

Topology:



Step 1: Configure loopbacks and assign addresses.

R1 Console

```
enable
conf t
hostname R1
interface Lo1
ip address 192.168.1.1 255.255.255.0
exit
interface s1/0
ip address 172.16.12.1 255.255.255.248
no shutdown
exit
interface s1/1
ip address 172.16.13.1 255.255.255.248
no shutdown
exit
router eigrp 100
network 192.168.1.0
network 172.16.12.0
network 172.16.13.0
no auto-summary
exit
R1#sh ip eigrp neighbors
R1#sh ip route
```

```

R1#
R1#sh ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address                Interface      Hold Uptime    SRTT    RTO  Q  Seq
                               (sec)          (ms)          Cnt  Num
1   172.16.13.3             Ser1/1        11 00:23:17    33    200  0  33
0   172.16.12.2             Ser1/0        11 00:58:47    46    276  0  27
R1#
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/29 is subnetted, 4 subnets
D       172.16.34.0 [90/2681856] via 172.16.13.3, 00:18:00, Serial1/1
D       172.16.23.0 [90/2681856] via 172.16.13.3, 00:23:25, Serial1/1
        [90/2681856] via 172.16.12.2, 00:23:25, Serial1/0
C       172.16.12.0 is directly connected, Serial1/0
C       172.16.13.0 is directly connected, Serial1/1
    192.168.4.0/25 is subnetted, 2 subnets
D       192.168.4.0 [90/2809856] via 172.16.13.3, 00:16:48, Serial1/1
D       192.168.4.128 [90/2809856] via 172.16.13.3, 00:16:48, Serial1/1
C       192.168.1.0/24 is directly connected, Loopback1
D       192.168.2.0/24 [90/2297856] via 172.16.12.2, 00:23:25, Serial1/0
D       192.168.3.0/24 [90/2297856] via 172.16.13.3, 00:23:25, Serial1/1
R1#

```

R2 Console

Router>enable

Router#conf t

Router(config)#hostname R2

R2(config)#interface Lo2

R2(config-if)#ip address 192.168.2.1 255.255.255.0

R2(config-if)#exit

R2(config)#interface s1/0

R2(config-if)#ip address 172.16.12.2 255.255.255.248

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#interface s1/1

R2(config-if)#ip address 172.16.23.2 255.255.255.248

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#router eigrp 100

R2(config-router)#network 192.168.2.0

R2(config-router)#network 172.16.12.0

R2(config-router)#network 172.16.23.0

R2(config-router)#no auto-summary

R2#sh ip eigrp neighbors

```

R2#sh ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT    RTO  Q  Seq
                               (sec)          (ms)          Cnt  Num
1   172.16.23.3         Se1/1         14 00:57:08    34    204  0   34
0   172.16.12.1         Se1/0         12 01:01:17    46    276  0   33
R2#
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/29 is subnetted, 4 subnets
D       172.16.34.0 [90/2681856] via 172.16.23.3, 00:20:40, Serial1/1
C       172.16.23.0 is directly connected, Serial1/1
C       172.16.12.0 is directly connected, Serial1/0
D       172.16.13.0 [90/2681856] via 172.16.23.3, 00:57:25, Serial1/1
        [90/2681856] via 172.16.12.1, 00:57:25, Serial1/0
    192.168.4.0/25 is subnetted, 2 subnets
D       192.168.4.0 [90/2809856] via 172.16.23.3, 00:19:28, Serial1/1
D       192.168.4.128 [90/2809856] via 172.16.23.3, 00:19:28, Serial1/1
D       192.168.1.0/24 [90/2297856] via 172.16.12.1, 00:26:05, Serial1/0
C       192.168.2.0/24 is directly connected, Loopback2
D       192.168.3.0/24 [90/2297856] via 172.16.23.3, 00:26:05, Serial1/1
R2#

```

R3 Console

```

Router>enable
Router#conf t
Router(config)#hostname R3
R3(config)#interface Lo3
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#exit
R3(config)#interface s1/0
R3(config-if)#ip address 172.16.13.3 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface s1/1
R3(config-if)#ip address 172.16.23.3 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface s1/2
R3(config-if)#ip address 172.16.34.3 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#router eigrp 100
R3(config-router)#network 192.168.3.0
R3(config-router)#network 172.16.13.0
R3(config-router)#network 172.16.23.0
R3(config-router)#network 172.16.34.0
R3(config-router)#no auto-summary
R3#sh ip eigrp neighbors

```

```

R3
Connected to Dynamips VM "R3" (ID 2, type c7200) - Console port
Press ENTER to get the prompt.

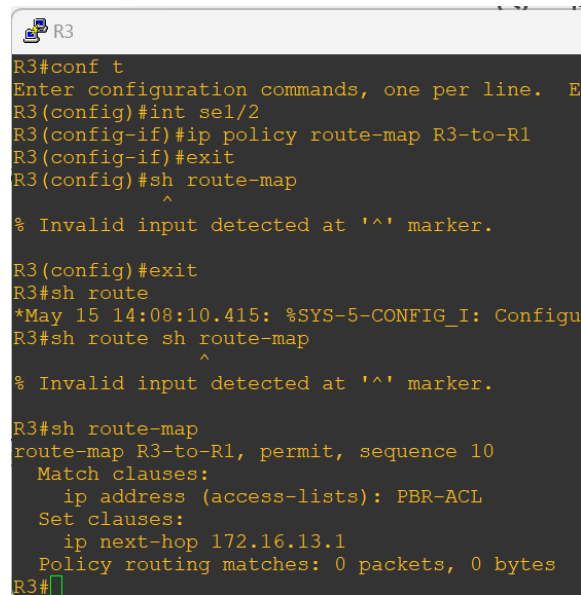
R3(config)#exit
R3#
*May 15 14:42:55.407: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address           Interface      Hold Uptime    SRTT    RTO  Q  Seq
                               (sec)          (ms)          Cnt  Num
2   172.16.34.4         Se1/2         12 00:21:08    59    354  0   7
1   172.16.13.1         Se1/0         11 00:27:29    45    270  0   34
0   172.16.23.2         Se1/1         13 00:58:50    55    330  0   28
R3#

```

```

R3(config)#ip access-list standard PBR-ACL
R3(config-std-nacl)#remark ACL matches R4 LAN B traffic
R3(config-std-nacl)#permit 192.168.4.128 0.0.0.127
R3(config-std-nacl)#exit
R3(config)#route-map R3-to-R1 permit
R3(config-route-map)#match ip address PBR-ACL
R3(config-route-map)#set ip next-hop 172.16.13.1
R3(config-route-map)#end
R3(config)#int s1/2
R3(config-if)#ip policy route-map R3-to-R1
R3(config-if)#exit
R3#sh route-map

```



```

R3#conf t
Enter configuration commands, one per line. E
R3(config)#int s1/2
R3(config-if)#ip policy route-map R3-to-R1
R3(config-if)#exit
R3(config)#sh route-map
^
% Invalid input detected at '^' marker.

R3(config)#exit
R3#sh route
*May 15 14:08:10.415: %SYS-5-CONFIG_I: Configu
R3#sh route sh route-map
^
% Invalid input detected at '^' marker.

R3#sh route-map
route-map R3-to-R1, permit, sequence 10
  Match clauses:
    ip address (access-lists): PBR-ACL
  Set clauses:
    ip next-hop 172.16.13.1
  Policy routing matches: 0 packets, 0 bytes
R3#

```

```

R3(config)#access-list 1 permit 192.168.4.0 0.0.0.255

```

R4 Console

```

Router>enable
Router#conf t
Router(config)#hostname R4
R4(config)#interface lo4
R4(config-if)#ip address 192.168.4.1 255.255.255.128
R4(config-if)#exit
R4(config)#interface lo5
R4(config-if)#ip address 192.168.4.129 255.255.255.128
R4(config-if)#exit
R4(config)#interface s1/0
R4(config-if)#ip address 172.16.34.4 255.255.255.248
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#router eigrp 100
R4(config-router)#network 192.168.4.0
R4(config-router)#network 172.16.34.0
R4(config-router)#no auto-summary
R4#sh ip eigrp neighbors

```

```
R4#sh ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address                Interface    Hold Uptime    SRTT  RTO  Q  Seq
                               (sec)          (ms)        354  0  37
0   172.16.34.3             Ser1/0      12 00:01:06    59
R4#
```

Before Route Maps

R4#traceroute 192.168.1.1 source 192.168.4.1

```
R4#traceroute 192.168.1.1 source 192.168.4.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1

 1 172.16.34.3 176 msec 164 msec 32 msec
 2 172.16.13.1 92 msec 156 msec 64 msec
```

R4#traceroute 192.168.1.1 source 192.168.4.129

```
R4#traceroute 192.168.1.1 source 192.168.4.129
Type escape sequence to abort.
Tracing the route to 192.168.1.1

 1 172.16.34.3 20 msec 52 msec 24 msec
 2 172.16.13.1 72 msec 52 msec 64 msec
```

After Route Maps

R4#traceroute 192.168.1.1 source 192.168.4.1

R4#traceroute 192.168.1.1 source 192.168.4.129

```
R4#traceroute 192.168.1.1 source 192.168.4.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1

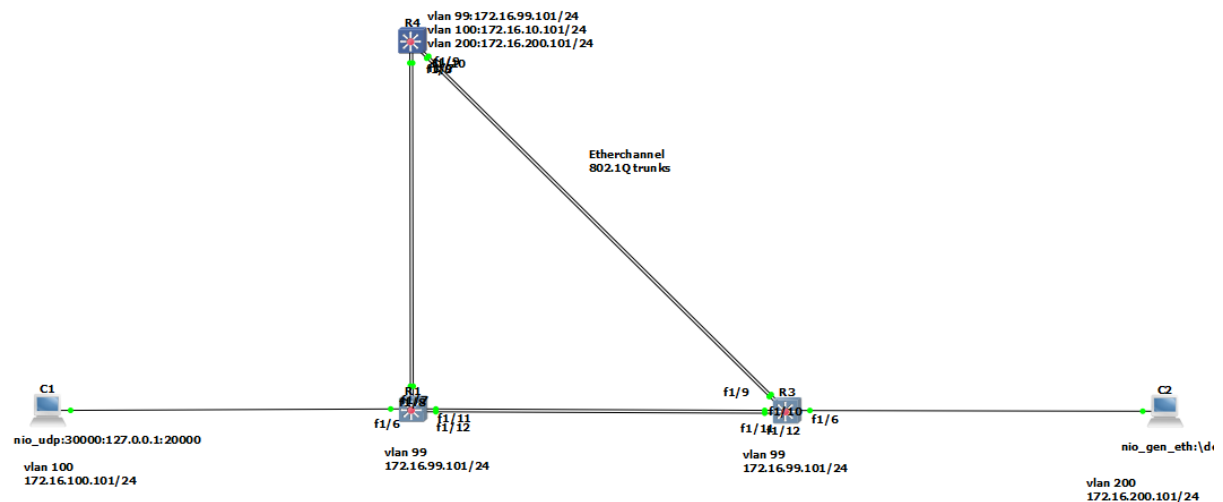
 1 172.16.34.3 40 msec 24 msec 36 msec
 2 172.16.13.1 60 msec 64 msec 68 msec
R4#
R4#traceroute 192.168.1.1 source 192.168.4.129
Type escape sequence to abort.
Tracing the route to 192.168.1.1

 1 172.16.34.3 20 msec 28 msec 32 msec
 2 172.16.13.1 64 msec 64 msec 56 msec
```

PRACTICAL 6

AIM: To Simulate IP Service Level Agreements and Remote SPAN in a Campus Environment

Topology:



Step 1: Configure loopbacks and assign addresses.

DLS1

DLS1# conf t

Enter configuration commands, one per line. End with CNTL/Z.

DLS1(config)# interface vlan 99

DLS1(config-if)# ip address 172.16.99.1 255.255.255.0

DLS1(config-if)# no shutdown

DLS1(config-if)# exit

DLS1(config)# enable secret class

DLS1(config)# line vty 0 15

DLS1(config-line)# no login

DLS1(config-line)# privilege level 15

```
Router>en
Router#conf t
Enter configuration commands, one per line. End
Router(config)#hostname DLS1
DLS1(config)#int vlan 99
DLS1(config-if)#ip add 172.16.99.1 255.255.255.0
DLS1(config-if)#no shut
DLS1(config-if)#exit
DLS1(config)#enable secret class
DLS1(config)#line vty 0 15
DLS1(config-line)#no login
DLS1(config-line)#privilege level 15
DLS1(config-line)#
```

ALS1

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname ALS1

ALS1(config)#interface vlan 99

ALS1(config-if)#ip add 172.16.99.1 255.255.255.0

ALS1(config-if)#no shut

ALS1(config-if)#exit

```
ALS1(config)#enable secret class
ALS1(config)#line vty 0 15
ALS1(config-line)#no login
ALS1(config-line)#privilege level 15
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ALS1
ALS1(config)#interface vlan 99
ALS1(config-if)#ip add 172.16.99.1 255.255.255.0
ALS1(config-if)#no shut
ALS1(config-if)#exit
ALS1(config)#enable secret class
ALS1(config)#line vty 0 15
ALS1(config-line)#no login
ALS1(config-line)#privilege level 15
```

ALS2

```
ALS2# conf t
Enter configuration commands, one per line. End with CNTL/Z.
ALS2(config)# interface vlan 99
ALS2(config-if)# ip address 172.16.99.1 255.255.255.0
ALS2(config-if)# no shutdown
ALS2(config-if)# exit
ALS2# conf t
ALS2(config)# enable secret class
ALS2(config)# line vty 0 15
ALS2(config-line)# no login
ALS2(config-line)# privilege level 15
```

```
ALS2(config)#int vlan 99
ALS2(config-if)#ip add 172.16.99.1 255.255.255.0
ALS2(config-if)#no shut
ALS2(config-if)#exit
ALS2(config)#enable secret class
ALS2(config)#line vty 0 15
ALS2(config-line)#no login
ALS2(config-line)#privilege level 15
ALS2(config-line)#
```

```
ALS1(config)# ip default-gateway 172.16.99.1
ALS2(config)# ip default-gateway 172.16.99.1
```

Step 2: Configure host PCs.

host A > ip 172.16.100.101/24 172.16.100.1

host B > ip 172.16.200.101/24 172.16.200.1

Step 3: Configure trunks and EtherChannels between switches

Configure the trunks and EtherChannel from DLS1 to ALS1

```
DLS1(config-line)# exit
DLS1(config)# int ran fastethernet 1/7 - 8
DLS1(config-if-range)# switchport trunk encapsulation
dot1q
DLS1(config-if-range)# switchport mode trunk
DLS1(config-if-range)# channel-group 1 mode on
DLS1(config-if-range)# no shut
DLS1(config-if-range)# exit
```


Configure the trunks and EtherChannel from DLS1 to ALS2

```
DLS1(config)# int ran fastethernet 1/9 - 10
DLS1(config-if-range)# switchport trunk encapsulation dot1q
DLS1(config-if-range)# switchport mode trunk
DLS1(config-if-range)# channel-group 2 mode on
DLS1(config-if-range)# no shut
```

Configure the trunks and EtherChannel from ALS1 and DLS1

```
ALS1# conf t
ALS1(config)# int ran fastethernet 1/11 - 12
ALS1(config-if-range)# switchport mode trunk
ALS1(config-if-range)# channel-group 3 mode on
ALS1(config-if-range)# no shut
```

Configure the trunks and EtherChannel and ALS1 and ALS2

```
ALS1(config-if-range)# exit
ALS1(config)# interface range fastethernet 1/7 – 8
ALS1(config-if-range)# switchport mode trunk
ALS1(config-if-range)# channel-group 2 mode on
ALS1(config-if-range)# no shut
```

Configure the trunks and EtherChannel from ALS2 and DLS1

```
ALS2# conf t
ALS2(config)# int ran fastethernet 1/11 – 12
ALS2(config-if-range)# switchport mode trunk
ALS2(config-if-range)# channel-group 3 mode on
ALS2(config-if-range)# no shut
```

Configure the trunks and EtherChannel and ALS2 and ALS1

```
ALS2(config-if-range)# exit
ALS2(config)# int ran fastethernet 1/9 – 10
ALS2(config-if-range)# switchport mode trunk
ALS2(config-if-range)# channel-group 2 mode on
ALS2(config-if-range)# no shut
```

Step 4: Configure VTP on ALS1 and ALS2

```
ALS1(config)# vtp mode client
```

```
ALS2(config)# vtp mode client
```

Step 5: Configure VTP on DLS1

```
DLS1(config)# vtp domain SWPOD
DLS1(config)# vtp version 2
DLS1(config)# vlan 99
DLS1(config-vlan)# name Management
DLS1(config-vlan)# vlan 100
DLS1(config-vlan)# name Finance
DLS1(config-vlan)# vlan 200
DLS1(config-vlan)# name Engineering
DLS1(config-vlan)# exit
```

Step 6: Configure access ports.

```
ALS1(config)# interface fastEthernet 1/6
ALS1(config-if)# switchport mode access
ALS1(config-if)# switchport access vlan 100
ALS1(config-if)# no shut
```

```
ALS2(config)# interface fastEthernet 1/6
ALS2(config-if)# switchport mode access
ALS2(config-if)# switchport access vlan 200
ALS1(config-if)# no shut
```

Step 7: Configure VLAN interfaces and enable routing

```
DLS1(config)# interface vlan 100
DLS1(config-if)# ip address 172.16.100.1 255.255.255.0
DLS1(config-if)# interface vlan 200
DLS1(config-if)# ip address 172.16.200.1 255.255.255.0
DLS1(config-if)# exit
DLS1(config)# ip routing
DLS1(config)# exit
DLS1# show ip route
DLS1# tclsh
foreach address {
172.16.99.1
172.16.99.101
172.16.99.102
172.16.100.1
172.16.200.1
172.16.100.101
172.16.200.101
} {
ping $address }
```

Part 2: Configure Cisco IOS IP SLAStep 1: Configure Cisco IOS IP SLA responders**ALS1**

```
ALS1(config)# exit
ALS1(config)# ip sla responder
ALS1(config)# ip sla responder udp-echo ipaddress
172.16.99.1 port 5000
```

ALS2

```
ALS2(config)# exit
ALS2(config)# ip sla responder
ALS2(config)# ip sla responder udp-echo ipaddress
172.16.99.1 port 5000
```

Step 2: Configure the Cisco IOS IP SLA source to measure network performance**DLS1**

```
DLS1(config)# ip sla 1
DLS1(config-ip-sla)# icmp-echo 172.16.100.101
DLS1(config-ip-sla-echo)# exit
DLS1(config)# ip sla 2
```

```
DLS1(config-ip-sla)# icmp-echo 172.16.200.101
DLS1(config-ip-sla-echo)# exit
DLS1(config)# ip sla 3
DLS1(config-ip-sla)# udp-jitter 172.16.99.101 5000
DLS1(config-ip-sla-jitter)# exit
DLS1(config)# ip sla 4
DLS1(config-ip-sla)# udp-jitter 172.16.99.102 5000
DLS1(config-ip-sla-jitter)# exit
```

```
DLS1(config)# ip sla schedule 1 life forever start-time now
DLS1(config)# ip sla schedule 2 life forever start-time now
DLS1(config)# ip sla schedule 3 life forever start-time now
DLS1(config)# ip sla schedule 4 life forever start-time now
DLS1(config)# exit
```

Step 3: Monitor IP SLAs operations

DLS1

```
DLS1# show ip sla configuration 1
DLS1# show ip sla configuration 3
DLS1# show ip sla application
```

ALS1

```
ALS1# show ip sla responder
```

DLS1

```
DLS1# show ip sla statistics 1
DLS1# show ip sla statistics 3
```

ALS1

```
ALS1(config)# interface vlan 99
ALS1(config-if)# shutdown
```

DLS1

```
DLS1# show ip sla statistics 3
```

Part 3: Switch Port Analyzer (SPAN) Feature

Step 1: Configure Remote SPAN (RSPAN)

DLS1

```
DLS1(config)# vlan 300
DLS1(config-vlan)# name REMOTE_SPAN
DLS1(config-vlan)# remote-span
DLS1# show vlan brief | include active
```

ALS1

```
ALS1# show vlan brief | include active
ALS1(config)# monitor session 1 source interface Fa0/6
ALS1(config)# monitor session 1 destination remote vlan300
ALS1# show monitor
```

ALS2

```
ALS2(config)# monitor session 10 source remote vlan 300
ALS2(config)# monitor session 10 destination interface Fa0/6
ALS2# show monitor
```

ALS2# show interface f0/6

Step 2: Test RSPAN operation

Step 3: End of Lab

DLS1 :

```
DLS1# show run brief | exclude !
hostname DLS1
boot-start-marker
boot-end-marker
enable secret 5 $1$394h$4u5EACpRNVCKAFz1Ntit6/
no aaa new-model
clock timezone CST -6 0
clock summer-time CDT recurring
system mtu routing 1500
ip routing
no ip domain-lookup
ip domain-name CCNP.NET
spanning-tree mode pvst
spanning-tree extend system-id
vlan internal allocation policy ascending
interface Port-channel1
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
interface Port-channel2
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
interface FastEthernet0/1
shutdown
interface FastEthernet0/2
shutdown
interface FastEthernet0/3
shutdown
interface FastEthernet0/4
shutdown
interface FastEthernet0/5
shutdown
interface FastEthernet0/6
shutdown
interface FastEthernet0/7
description EtherChannel to ALS1
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
channel-group 1 mode active
interface FastEthernet0/8
description EtherChannel to ALS1
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
```

```
channel-group 1 mode active
interface FastEthernet0/9
description EtherChannel to ALS2
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
channel-group 2 mode active
interface FastEthernet0/10
description EtherChannel to ALS2
switchport trunk encapsulation dot1q
switchport trunk native vlan 666
switchport mode trunk
channel-group 2 mode active
interface FastEthernet0/11
shutdown
interface FastEthernet0/12
shutdown
interface FastEthernet0/13
shutdown
interface FastEthernet0/14
shutdown
interface FastEthernet0/15
shutdown
interface FastEthernet0/16
shutdown
interface FastEthernet0/17
shutdown
interface FastEthernet0/18
shutdown
interface FastEthernet0/19
shutdown
interface FastEthernet0/20
shutdown
interface FastEthernet0/21
shutdown
interface FastEthernet0/22
shutdown
interface FastEthernet0/23
shutdown
interface FastEthernet0/24
shutdown
interface GigabitEthernet0/1
shutdown
interface GigabitEthernet0/2
shutdown
interface Vlan1
no ip address
interface Vlan99
ip address 172.16.99.1 255.255.255.0
interface Vlan100
ip address 172.16.100.1 255.255.255.0
interface Vlan200
```

```
ip address 172.16.200.1 255.255.255.0
ip http server
ip http secure-server
ip sla 1
icmp-echo 172.16.100.101
ip sla schedule 1 life forever start-time now
ip sla 2
icmp-echo 172.16.200.101
ip sla schedule 2 life forever start-time now
ip sla 3
udp-jitter 172.16.99.101 5000
ip sla schedule 3 life forever start-time now
ip sla 4
udp-jitter 172.16.99.102 5000
ip sla schedule 4 life forever start-time now
line con 0
exec-timeout 0 0
logging synchronous
line vty 0 4
password cisco
login
line vty 5 15
password cisco
login
end
DLS1#
```

ALS1:

```
ALS1# show run brief | exclude !
Building configuration.
Current configuration : 2692 bytes
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
hostname ALS1
boot-start-marker
boot-end-marker
enable secret 5 $1$eOw5$e1Pah.syL1h.EPTDwlpnG0
no aaa new-model
clock timezone CST -6 0
clock summer-time CDT recurring
system mtu routing 1500
no ip domain-lookup
ip domain-name CCNP.NET
spanning-tree mode pvst
spanning-tree extend system-id
vlan internal allocation policy ascending
interface Port-channel1
switchport trunk native vlan 666
```

```
switchport mode trunk
interface Port-channel3
switchport trunk native vlan 666
switchport mode trunk
interface FastEthernet0/1
shutdown
interface FastEthernet0/2
shutdown
interface FastEthernet0/3
shutdown
interface FastEthernet0/4
shutdown
interface FastEthernet0/5
shutdown
interface FastEthernet0/6
switchport access vlan 100
switchport mode access
interface FastEthernet0/7
description EtherChannel to DLS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 1 mode active
interface FastEthernet0/8
description EtherChannel to DLS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 1 mode active
interface FastEthernet0/9
shutdown
interface FastEthernet0/10
shutdown
interface FastEthernet0/11
description EtherChannel to ALS2
switchport trunk native vlan 666
switchport mode trunk
channel-group 3 mode active
interface FastEthernet0/12
description EtherChannel to ALS2
switchport trunk native vlan 666
switchport mode trunk
channel-group 3 mode active
interface FastEthernet0/13
shutdown
interface FastEthernet0/14
shutdown
interface FastEthernet0/15
shutdown
interface FastEthernet0/16
shutdown
interface FastEthernet0/17
shutdown
interface FastEthernet0/18
```

```
shutdown
interface FastEthernet0/19
shutdown
interface FastEthernet0/20
shutdown
interface FastEthernet0/21
shutdown
interface FastEthernet0/22
shutdown
interface FastEthernet0/23
shutdown
interface FastEthernet0/24
shutdown
interface GigabitEthernet0/1
shutdown
interface GigabitEthernet0/2
shutdown
interface Vlan1
no ip address
interface Vlan99
ip address 172.16.99.101 255.255.255.0
ip default-gateway 172.16.99.1
ip http server
ip http secure-server
ip sla responder
ip sla responder udp-echo ipaddress 172.16.99.1 port 5000
line con 0
exec-timeout 0 0
logging synchronous
line vty 0 4
password cisco
login
line vty 5 15
password cisco
login
monitor session 1 source interface Fa0/6
monitor session 1 destination remote vlan 300
end
ALS1#
```

ALS2:

```
ALS2# show run brief | exclude !
hostname ALS2
boot-start-marker
boot-end-marker
enable secret 5 $1$QTDK$sDfAMuL5FIHWCO5U4q9F50
no aaa new-model
clock timezone CST -6 0
clock summer-time CDT recurring
system mtu routing 1500
no ip domain-lookup
ip domain-name CCNP.NET
```



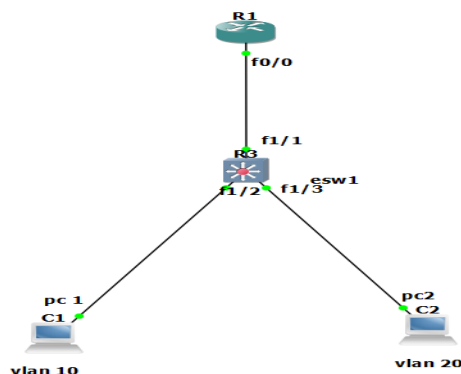
```
spanning-tree mode pvst
spanning-tree extend system-id
vlan internal allocation policy ascending
interface Port-channel2
switchport trunk native vlan 666
switchport mode trunk
interface Port-channel3
switchport trunk native vlan 666
switchport mode trunk
interface FastEthernet0/1
shutdown
interface FastEthernet0/2
shutdown
interface FastEthernet0/3
shutdown
interface FastEthernet0/4
shutdown
interface FastEthernet0/5
shutdown
interface FastEthernet0/6
switchport access vlan 200
switchport mode access
interface FastEthernet0/7
shutdown
interface FastEthernet0/8
shutdown
interface FastEthernet0/9
description EtherChannel to DLS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 2 mode active
interface FastEthernet0/10
description EtherChannel to DLS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 2 mode active
interface FastEthernet0/11
description EtherChannel to ALS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 3 mode active
interface FastEthernet0/12
description EtherChannel to ALS1
switchport trunk native vlan 666
switchport mode trunk
channel-group 3 mode active
interface FastEthernet0/13
shutdown
interface FastEthernet0/14
shutdown
interface FastEthernet0/15
shutdown
```

```
interface FastEthernet0/16
shutdown
interface FastEthernet0/17
shutdown
interface FastEthernet0/18
shutdown
interface FastEthernet0/19
shutdown
interface FastEthernet0/20
shutdown
interface FastEthernet0/21
shutdown
interface FastEthernet0/22
shutdown
interface FastEthernet0/23
shutdown
interface FastEthernet0/24
shutdown
interface GigabitEthernet0/1
shutdown
interface GigabitEthernet0/2
shutdown
interface Vlan1
no ip address
interface Vlan99
ip address 172.16.99.102 255.255.255.0
ip default-gateway 172.16.99.1
ip http server
ip http secure-server
ip sla responder
ip sla responder udp-echo ipaddress 172.16.99.1 port 5000
line con 0
exec-timeout 0 0
logging synchronous
line vty 0 4
password cisco
login
line vty 5 15
password cisco
login
monitor session 10 destination interface Fa0/6
monitor session 10 source remote vlan 300
end
ALS2#
```

PRACTICAL 7

AIM: Inter-VLAN Routing.

NETWORK TOPOLOGY:



Code and Output:

Configuration for Router R1:

```

R1#conf t
R1(config)#int f 0/0
R1(config-if)#no ip address
R1(config-if)#no shut
R1(config-if)#int f 0/0.1
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 10.0.0.1 255.255.255.0
R1(config-subif)#exit
R1(config)#int f 0/0.2
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 20.0.0.1 255.255.255.0
R1(config-subif)#exit
  
```

```

R1#en
R1#conf t
Enter configuration commands, one per line. End
R1(config)#int f0/0
R1(config-if)#no ip add
R1(config-if)#no shut
R1(config-if)#
*May 17 14:12:39.563: %LINK-3-UPDOWN: Interface Fa
changed state to up
R1(config-if)#
*May 17 14:12:39.563: %ENTITY_ALARM-6-INFO: CLEAR
cal Port Administrative State_Down
*May 17 14:12:40.563: %LINEPROTO-5-UPDOWN: Line p
ace FastEthernet0/0, changed state to up
R1(config-if)#int f0/0.1
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip add 10.0.0.1 255.255.255.0
R1(config-subif)#exit
R1(config)#int f 0/0.2
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip add 20.0.0.1 255.255.255.0
R1(config-subif)#exit
  
```

Configuration for Switch:

```

ESW1#
ESW1#vlan database
ESW1(vlan)#vlan 10 name IT
ESW1(vlan)#vlan 20 name HR
ESW1(vlan)#exit
ESW1#conf t
ESW1(config)#int f 1/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#int f 1/3
  
```

```
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 20
ESW1(config-if)#exit
ESW1(config)#int f 1/1
ESW1(config-if)#switch port mode trunk
ESW1(config-if)#switchport trunk encapsulation dot1Q
ESW1(config-if)#exit
ESW1(config)#
```

Configuration for PC1

```
PC1>
PC1>ip 10.0.0.2 /24 10.0.0.1
PC1>save
PC1>ping 10.0.0.1
PC1>
```

Configuration for PC2

```
PC2>
PC2>ip 20.0.0.2 /24 20.0.0.1
PC2>save
PC2>ping 20.0.0.1
PC2>
```

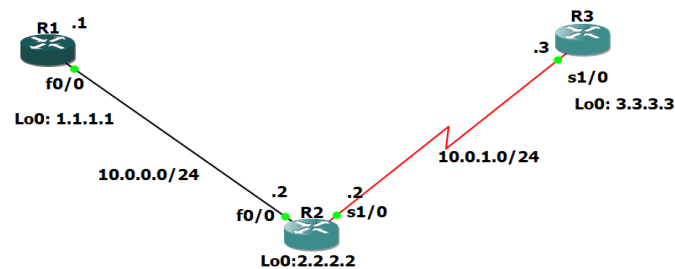
RESULT:

```
ESW1#show vlan-switch
```

PRACTICAL 8

AIM: Simulating MPLS environment

NETWORK TOPOLOGY:



Code and Output:

Step 1 – IP addressing of MPLS Core and OSPF

First bring 3 routers into your topology R1, R2, R3 position them as below. We are going to address the routers and configure ospf to ensure loopback to loopback connectivity between R1 and R3

```

R1
Press ENTER to get the prompt.

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int lo0
R1(config-if)#ip add
*Jun 20 13:44:45.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R1(config-if)#ip add 1.1.1.1 255.255.255.255
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int f0/0
R1(config-if)#ip add 10.0.0.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip ospf
*Jun 20 13:45:35.743: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R1(config-if)#ip ospf
*Jun 20 13:45:36.743: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port Administrative State Down
*Jun 20 13:45:36.743: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#ip ospf 1 area 0
R1(config-if)#

R2(config-if)#ip ospf
*Jun 20 14:06:29.299: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port Administrative State Down
R2(config-if)#ip ospf
*Jun 20 14:06:30.295: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#ip ospf 1 area 0
R2(config-if)#
R2(config-if)#int
*Jun 20 14:08:45.535: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R2(config-if)#int s1/0
R2(config-if)#ip add 10.0.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#ip ospf
*Jun 20 14:13:56.775: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R2(config-if)#ip ospf a
*Jun 20 14:13:56.775: %ENTITY_ALARM-6-INFO: CLEAR INFO Ser1/0 Physical Port Administrative State Down
R2(config-if)#ip ospf a
*Jun 20 14:13:57.779: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R2(config-if)#ip ospf 1 area 0
R2(config-if)#
  
```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#hostname R3
R3(config)#int lo0
R3(config-if)#ip add
*Jun 20 14:16:05.607: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Loopback0, changed state to up
R3(config-if)#ip add 3.3.3.3 255.255.255.255
R3(config-if)#ip ospf 1 area 0
R3(config-if)#int s1/0
R3(config-if)#ip add 10.0.1.3 255.255.255.0
R3(config-if)#no shut
R3(config-if)#ip ospf
*Jun 20 14:16:57.559: %LINK-3-UPDOWN: Interface Serial1/0, changed st
ate to up
R3(config-if)#ip ospf 1 area
*Jun 20 14:16:57.559: %ENTITY_ALARM-6-INFO: CLEAR INFO Ser1/0 Physical
Port Administrative State Down
*Jun 20 14:16:58.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial1/0, changed state to up
R3(config-if)#ip ospf 1 area 0
R3(config-if)#
*Jun 20 14:17:06.399: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Seria
l1/0 from LOADING to FULL, Loading Done
R3(config-if)#

```

You should now have full ip connectivity between R1, R2, R3 to verify this we need to see if we can ping between the loopbacks of R1 and R3

R1#ping 3.3.3.3 source lo0

```

Connected to Dynamips VM "R1" (ID 0, type c7200) - Console port
Press ENTER to get the prompt.

R1#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/74/88 ms

R3#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/70/80 ms

```

You could show the routing table here, but the fact that you can ping between the loopbacks is verification enough and it is safe to move on.

Step 2 - Configure LDP on all the interfaces in the MPLS Core

In order to run MPLS you need to enable it, there are two ways to do this.

- At each interface enter the mpls ip command.
- Under the ospf process use the mpls ldp autoconfig command.

we will be using the second option, so go into the ospf process and enter mpls ldp autoconfig - this will enable mpls label distribution protocol on every interface running ospf under that specific process.

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#mpls ldp aut
R1(config-router)#mpls ldp autoconfig
R1(config-router)#

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#mpls ldp autoconfig
R2(config-router)#
*Jun 20 14:33:24.719: %LDP-5-NBRCHG: LDP Neighbor 1.1.1.1:0 (1) is UP

```

```

R3
Connected to Dynamips VM "R3" (ID 2, type c7200) - Console port
Press ENTER to get the prompt.

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#mpls ldp autoconfig
R3(config-router)#
*Jun 20 14:35:43.843: %LDP-5-NBRCHG: LDP Neighbor 2.2.2.2:0 (1) i
s UP

```

To verify the mpls interfaces the command is very simple - sh mpls interface.

This is done on R2 and you can see that both interfaces are running mpls and using LDP.

```

R2#sh mpls interface
Interface          IP          Tunnel    Operational
FastEthernet0/0    Yes (ldp)   No        Yes
Serial1/0          Yes (ldp)   No        Yes
R2#

```

You can also verify the LDP neighbors with the sh mpls ldp neighbors command.

```

R2#sh mpls ldp neigh
Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 2.2.2.2:0
TCP connection: 1.1.1.1.646 - 2.2.2.2.51127
State: Oper; Msgs sent/rcvd: 15/15; Downstream
Up time: 00:06:22
LDP discovery sources:
FastEthernet0/0, Src IP addr: 10.0.0.1
Addresses bound to peer LDP Ident:
10.0.0.1 1.1.1.1
Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 2.2.2.2:0
TCP connection: 3.3.3.3.11398 - 2.2.2.2.646
State: Oper; Msgs sent/rcvd: 12/12; Downstream
Up time: 00:04:03
LDP discovery sources:
Serial1/0, Src IP addr: 10.0.1.3
Addresses bound to peer LDP Ident:
10.0.1.3 3.3.3.3

```

One more verification to confirm LDP is running ok is to do a trace between R1 and R3 and verify if you get MPLS Labels show up in the trace.

```

R1#trace 3.3.3.3

Type escape sequence to abort.
Tracing the route to 3.3.3.3

 0 10.0.0.2 [MPLS: Label 17 Exp 0] 64 msec 56 msec 68 msec
 1 10.0.1.3 64 msec 64 msec 72 msec

```

As you can see the trace to R2 used an MPLS Label in the path, as this is a very small MPLS core only one label was used as R3 was the final hop.

So to review we have now configured IP addresses on the MPLS core, enabled OSPF and full IP connectivity between all routers and finally enabled mpls on all the interfaces in the core and have established ldp neighbors between all routers.

The next step is to configure MP-BGP between R1 and R3

This is when you start to see the layer 3 vpn configuration come to life

Step 3 - MPLS BGP Configuration between R1 and R3

We need to establish a Multi Protocol BGP session between R1 and R3 this is done by configuring the vpnv4 address family as below.

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router bgp 1
R1(config-router)#neighbor 3.3.3.3 remote-as 1
R1(config-router)#neighbor 3.3.3.3 update-source Lo0
R1(config-router)#no auto-summary
R1(config-router)#!
R1(config-router)#address-family vpnv4
R1(config-router-af)#neighbor 3.3.3.3 activate
R1(config-router-af)#

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router bgp 1
R3(config-router)#neighbor 1.1.1.1 remote-as 1
R3(config-router)#neighbor 1.1.1.1 update-source Lo0
R3(config-router)#no auto-summary
R3(config-router)#!
R3(config-router)#address-family vpnv4
R3(config-router-af)#neighbor 1.1.1.1
*Jun 20 14:51:08.019: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up

```

You should see log messages showing the BGP sessions coming up.

To verify the BGP session between R1 and R3 issue the command sh bgp vpnv4 unicast all summary

```

R1#sh bgp vpnv4 unicast all summary
BGP router identifier 1.1.1.1, local AS number 1
BGP table version is 1, main routing table version 1

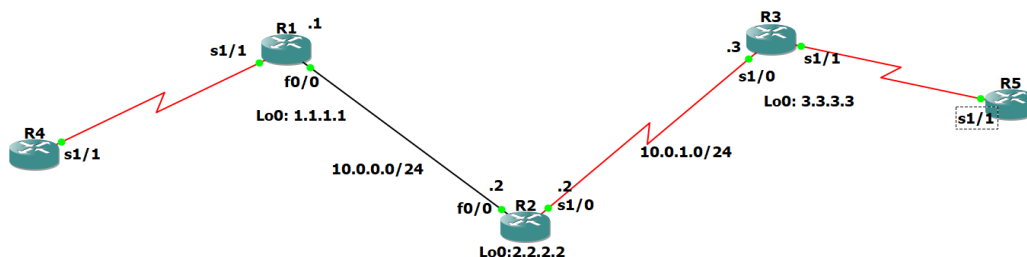
Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/
PfxRcd
3.3.3.3        4    1     13     13        1    0    0 00:07:16
0

```

You can see here that we do have a bgp vpnv4 peering to R3 - looking at the PfxRcd you can see it says 0 this is because we have not got any routes in BGP. We are now going to add two more routers to the topology. These will be the customer sites connected to R1 and R3. We will then create a VRF on each router and put the interfaces connected to each site router into that VRF.

Step 4 - Add two more routers, create VRFs

We will add two more routers into the topology so it now looks like the final topology.



Router 4 will peer OSPF using process number 2 to a VRF configured on R1. It will use the local site addressing of 192.168.1.0/24.

```

R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int lo0
R4(config-if)#ip add
*Jun 20 15:28:54.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R4(config-if)#ip add 4.4.4.4 255.255.255.255
R4(config-if)#ip ospf 2 area 2
R4(config-if)#int s1/1
R4(config-if)#ip add 192.168.1.4 255.255.255.0
R4(config-if)#ip ospf 2 area 2
R4(config-if)#no shut

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/1
R1(config-if)#no shut
R1(config-if)#ip add
*Jun 20 15:32:23.115: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
R1(config-if)#ip add
*Jun 20 15:32:23.115: %ENTITY_ALARM-6-INFO: CLEAR INFO Ser1/1 Physical Port Administrative State Down
R1(config-if)#ip add
*Jun 20 15:32:24.119: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R1(config-if)#ip add 192.168.1.1 255.255.255.0

```

Now at this point we have R4 peering to R1 but in the global routing table of R1 which is not what we want.

We are now going to start using VRF's

What is a VRF in networking?

Virtual routing and forwarding (VRF) is a technology included in IP (Internet Protocol) that allows multiple instances of a routing table to co-exist in a router and work together but not interfere with each other.. This increases functionality by allowing network paths to be segmented without using multiple devices.

As an example if R1 was a PE Provider Edge router of an ISP and it had two customers that were both addressed locally with the 192.168.1.0/24 address space it could accommodate both their routing tables in different VRFs - it distinguishes between the two of them using a Route Distinguisher

So back to the topology - we now need to create a VRF on R1

For this mpls I will be using VRF RED.

```
R1(config)#ip vrf RED
R1(config-vrf)#rd 4:4
R1(config-vrf)#route-target both 4:4
R1(config-vrf)#exit
R1(config)#int s1/1
R1(config-if)#ip vrf forwarding RED
% Interface Serial1/1 IP address 192.168.1.1 removed due to enabl
ing VRF RED
```

Now notice what happens when you do that - the IP address is removed which you will need to re-apply as follow.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/1
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#
```

Now if we view the config on R1 int f0/1 you can see the VRF configured.

```
R1#sh run int s1/1
Building configuration...

Current configuration : 112 bytes
!
interface Serial1/1
 ip vrf forwarding RED
 ip address 192.168.1.1 255.255.255.0
 serial restart-delay 0
end
```

Now we can start to look into VRF's and how they operate - you need to understand now that there are 2 routing tables within R1.

- The Global Routing Table
- The Routing Table for VRF RED

If you issue the command `sh ip route` this shows the routes in the global table and you will notice that you do not see 192.168.1.0/24

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
 2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.0.0.2, 01:26:46, FastEthernet0/0
 3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/66] via 10.0.0.2, 01:26:46, FastEthernet0/0
10.0.0.0/24 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
O       10.0.1.0 [110/65] via 10.0.0.2, 01:26:46, FastEthernet0/0
```

If you now issue the command `sh ip route vrf red` - this will show the routes in the routing table for VRF RED

```
R1#sh ip route vrf red
% IP routing table red does not exist
R1#
```

NOTE: The VRF name is case sensitive!

```
R1#sh ip route vrf RED

Routing Table: RED
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C      192.168.1.0/24 is directly connected, Serial1/1
R1#
```

We just need to enable OSPF on this interface and get the loopback address for R4 in the VRF RED routing table before proceeding.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int s1/1
R1(config-if)#ip ospf 2 area 2
R1(config-if)#
*Jun 20 15:47:32.423: %OSPF-5-ADJCHG: Process 2, Nbr 4.4.4.4 on Serial1/1 from LOADING to FULL, Loading Done
R1(config-if)#
```

If we now check the routes in the VRF RED routing table you should see 4.4.4.4 in there as well.

```
R1#sh ip route vrf RED

Routing Table: RED
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/65] via 192.168.1.4, 00:00:53, Serial1/1
C      192.168.1.0/24 is directly connected, Serial1/1
R1#
```

We now need to repeat this process for R3 & R6

Router 6 will peer OSPF using process number 2 to a VRF configured on R3. It will use the local site addressing of 192.168.2.0/24.

```
R5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#int lo0
R5(config-if)#ip add
*Jun 20 15:59:29.159: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R5(config-if)#ip add 6.6.6.6 255.255.255.255
R5(config-if)#ip ospf 2 area 2
R5(config-if)#int s1/1
R5(config-if)#ip add 192.168.2.6 255.255.255.0
R5(config-if)#ip ospf 2 area 2
R5(config-if)#no shut
```

```

R3(config)#int s1/1
R3(config-if)#no shut
R3(config-if)#ip add
*Jun 20 15:55:38.331: %LINK-3-UPDOWN: Interface Serial1/1, changed state to
up
R3(config-if)#ip add 192
*Jun 20 15:55:38.331: %ENTITY_ALARM-6-INFO: CLEAR INFO S1/1 Physical Port
Administrative State Down
R3(config-if)#ip add 192.16
*Jun 20 15:55:39.335: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial
1/1, changed state to up
R3(config-if)#ip add 192.168.2.3 255.255.255.0
R3(config-if)#

```

We also need to configure a VRF onto R3 as well.

```

R3(config)#ip vrf RED
R3(config-vrf)#rd 4:4
R3(config-vrf)#route-target both 4:4
R3(config-vrf)#

```

So now we have configured the VRF on R3 we need to move the interface F0/1 into that VRF

```

R3(config)#int s1/1
R3(config-if)#ip vrf forwarding RED
% Interface Serial1/1 IP address 192.168.2.3 removed due to enabling VRF RED
D

```

re-apply IP Address it.

```

R3(config-if)#int s1/1
R3(config-if)#ip add 192.168.2.1 255.255.255.0
R3(config-if)#

```

Now if we view the config on R3 int f0/1 you can see the VRF configured.

```

R3#sh run int s1/1
Building configuration...

Current configuration : 112 bytes
!
interface Serial1/1
 ip vrf forwarding RED
 ip address 192.168.2.1 255.255.255.0
 serial restart-delay 0
end

```

Finally we just need to enable OSPF on that interface and verify the routes are in the RED routing table.

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int s1/1
R3(config-if)#ip ospf 2 area 2

```

Check the routes in vrf RED.

```

R3#sh ip route vrf RED

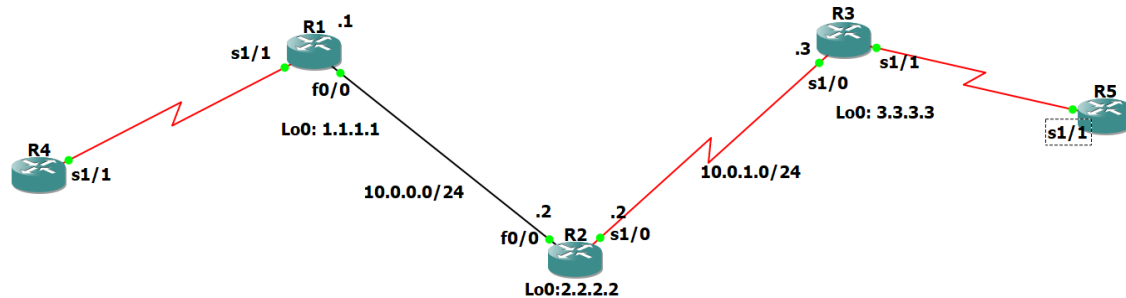
Routing Table: RED
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  6.0.0.0/32 is subnetted, 1 subnets
O       6.6.6.6 [110/65] via 192.168.2.6, 00:01:14, Serial1/1
C       192.168.2.0/24 is directly connected, Serial1/1

```

Ok so we have come a long way now let's review the current situation. We now have this setup



R1,R2,R3 form the MPLS Core and are running OSPF with all loopbacks running a /32 address and all have full connectivity. R1 and R3 are peering with MP-BGP. LDP is enabled on all the internal interfaces. The external interfaces of the MPLS core have been placed into a VRF called RED and then a site router has been joined to that VRF on each side of the MPLS core - (These represent a small office)

The final step to get full connectivity across the MPLS core is to redistribute the routes in OSPF on R1 and R3 into MP-BGP and MP-BGP into OSPF, this is what we are going to do now.

We need to redistribute the OSPF routes from R4 into BGP in the VRF on R1, the OSPF routes from R6 into MP-BGP in the VRF on R3 and then the routes in MP-BGP in R1 and R3 back out to OSPF.

let's do some verifications

Check the routes on R4

```
R4#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
C       192.168.1.0/24 is directly connected, Serial1/1
```

As expected we have the local interface and the loopback address.

When we are done we want to see 6.6.6.6 in there so we can ping across the MPLS

Check the routes on R1

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
    2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.0.0.2, 00:21:53, FastEthernet0/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/66] via 10.0.0.2, 00:21:53, FastEthernet0/0
    10.0.0.0/24 is subnetted, 2 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
O       10.0.1.0 [110/65] via 10.0.0.2, 00:21:53, FastEthernet0/0
```

Remember we have a VRF configured on this router so this command will show routes in the global routing table (the MPLS Core) and it will not show the 192.168.1.0/24 route as that is in VRF RED - to see that we run the following command.

```
R1#sh ip route vrf RED
Routing Table: RED
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    4.0.0.0/32 is subnetted, 1 subnets
O      4.4.4.4 [110/65] via 192.168.1.4, 00:26:39, Serial1/1
C      192.168.1.0/24 is directly connected, Serial1/1
R1#
```

Here you can see Routing Table: RED is shown and the routes to R4 are now visible with 4.4.4.4 being in OSPF.

So we need to do the following;

- Redistribute OSPF into MP-BGP on R1
- Redistribute MP-BGP into OSPF on R1
- Redistribute OSPF into MP-BGP on R3
- Redistribute MP-BGP into OSPF on R3

Redistribute OSPF into MP-BGP on R1

```
R1#conf t
Enter configuration commands, one per line.
R1(config)#router bgp 1
R1(config-router)#add-family ipv4 vrf RED
^
% Invalid input detected at '^' marker.

R1(config-router)#address-family ipv4 vrf RED
R1(config-router-af)#redistribute ospf 2
R1#
```

Redistribute OSPF into MP-BGP on R3

```
R3#conf t
Enter configuration commands, one per line.
R3(config)#router bgp 1
R3(config-router)#address-family ipv4 vrf RED
R3(config-router-af)#redistribute ospf 2
R3(config-router-af)#
R3#
```

This has enabled redistribution of the OSPF routes into BGP. We can check the routes from R4 and R6 are now showing in the BGP table for their VRF with this command

sh ip bgp vpnv4 vrf RED

```
R1#sh ip bgp vpnv4 vrf RED
BGP table version is 9, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 4:4 (default for vrf RED)
*> 4.4.4.4/32        192.168.1.4          65           32768 ?
*>i6.6.6.6/32        3.3.3.3              65          100      0 ?
*> 192.168.1.0       0.0.0.0              0           32768 ?
*>i192.168.2.0       3.3.3.3              0          100      0 ?
R1#
```

Here we can see that 4.4.4.4 is now in the BGP table in VRF RED on R1 with a next hop of 192.168.1.4 (R4) and also 6.6.6.6 is in there as well with a next hop of 3.3.3.3 (which is the loopback of R3 - showing that it is going over the MPLS and R1 is not in the picture)

The same should be true on R3

```

R3#sh ip bgp vpnv4 vrf RED
BGP table version is 9, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 4:4 (default for vrf RED)
*>i4.4.4.4/32        1.1.1.1             65      100      0 ?
*> 6.6.6.6/32        192.168.2.6         65              32768 ?
*>i192.168.1.0       1.1.1.1             0       100      0 ?
*> 192.168.2.0       0.0.0.0             0              32768 ?

```

Which it is! 6.6.6.6 is now in the BGP table in VRF RED on R3 with a next hop of 192.168.2.6 (R6) and also 4.4.4 is in there as well with a next hop of 1.1.1.1 (which is the loopback of R1 - showing that it is going over the MPLS and R2 is not in the picture)

The final step is to get the routes that have come across the MPLS back into OSPF and then we can get end to end connectivity

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 2
R1(config-router)#redistribute bgp 1 subnets
R1(config-router)#

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 2
R3(config-router)#redistribute bgp 1 subnets

```

If all has worked we should be now able to ping 6.6.6.6 from R4

Before we do let's see what the routing table looks like on R4

```

R4#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

   4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
   6.0.0.0/32 is subnetted, 1 subnets
O IA    6.6.6.6 [110/129] via 192.168.1.1, 00:02:32, Serial1/1
C       192.168.1.0/24 is directly connected, Serial1/1
O IA    192.168.2.0/24 [110/65] via 192.168.1.1, 00:02:32, Serial1/1

```

we have 6.6.6.6 in there

Also check the routing table on R5

```

R5#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

   4.0.0.0/32 is subnetted, 1 subnets
O IA    4.4.4.4 [110/129] via 192.168.2.1, 00:03:04, Serial1/1
   6.0.0.0/32 is subnetted, 1 subnets
C       6.6.6.6 is directly connected, Loopback0
O IA    192.168.1.0/24 [110/65] via 192.168.2.1, 00:03:04, Serial1/1
C       192.168.2.0/24 is directly connected, Serial1/1

```

we have 4.4.4.4 in there so we should be able to ping across the MPLS.

```
R4#ping 6.6.6.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 6.6.6.6, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 132/143/168 ms
```

Which we can - to prove this is going over the MPLS and be label switched and not routed, let's do a trace

```
R4#trace 6.6.6.6
Type escape sequence to abort.
Tracing the route to 6.6.6.6

 0  192.168.1.1  44 msec  36 msec  36 msec
 1  10.0.0.2    132 msec 128 msec 168 msec
 2  192.168.2.1  96 msec 120 msec 92 msec
 3  192.168.2.6 140 msec 140 msec 128 msec
R4#
```